Explicit Chevalley-Warning-with-symmetry search is PPA_p-complete for prime p

Mika Göös, Pritish Kamath,
Katerina Sotiraki, Manolis Zampetakis

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Editorial scope. One explicit ChevalleyWithSymmetry_p completeness mechanism with full membership/hardness,monomial/symmetry/labeled-CWT and required graph equivalence proofs. Only prime p. Ordinary Chevalley_p completeness remains unclaimed;SIS background is not upgraded or counted. Independent q classification not counted in this task.

Allowed standard prerequisites (editorial).

- Finite-field polynomial arithmetic,modulo-p counting,free cyclic actions and TFNP many-one reductions. Actual p is prime,q denotes an integer>=2. Standard finite-field/counting facts remain background;all author CWT/monomial/symmetry/labeled-degree and local graph/reduction proofs are retained. Full37 formal bibliography entries retained.

1.2 A Natural Complete Problem via Chevalley-Warning Theorem

There have been several works focusing on completeness results for the class PPA (i.e. PPA_2). Initial works showed the PPA-completeness of (non-natural) total search problems corresponding to topological fixed point theorems [24, 1, 19]. Closer to our paper, Belovs et al. [8] show the PPA-completeness of computational analogs of Combinatorial Nullstellensatz and the Chevalley–Warning Theorem, but which explicitly involve a circuit as part of the input. More recently, breakthrough results showed PPA-completeness of problems without a circuit or a Turing Machine in the input such as Consensus-Halving, Necklace-Splitting and Ham-Sandwich [20, 21] resolving an open problem since the definition of PPA in [33].

Our main contribution is to provide a natural complete problem for PPA_p, for every prime p; thereby also yielding a new complete problem for PPA. Our complete problem is an extension of the problem $\textsc{Chevalley}_p$, defined by Papadimitriou [33], which is a search problem associated to the celebrated Chevalley-Warning Theorem. We first present an abstract way to understand the proof of the Chevalley-Warning Theorem that motivates the definition of our natural complete problem for PPA_p.

1.2.1 Max-Degree Monic Monomials and Proof of Chevalley-Warning Theorem

In 1935, Claude Chevalley [15] resolved a hypothesis stated by Emil Artin, that all finite fields are quasi-algebraically closed. Later, Ewald Warning [37] proved a slight generalization of Chevalley's theorem. This generalized statement is usually referred to as the Chevalley-Warning Theorem (CWT, for short). Despite its initial algebraic motivation, CWT has found profound applications in combinatorics and number theory as we discuss in Section1.4 (and Section6).

We now explain the statement of the Chevalley-Warning Theorem, starting with some notations. For any field $\mathbb{F}$ and any polynomial f in a polynomial ring $\mathbb{F}[x_1, \ldots, x_n]$ we use $\deg(f)$ to represent the degree of f. We use $\boldsymbol{x}$ to succinctly denote the set of all variables $(x_1, \ldots, x_n)$ (the number of variables will always be n) and $\boldsymbol{f}$ to succinctly denote a system of polynomials $\boldsymbol{f} = (f_1, \ldots, f_m) \in \mathbb{F}[\boldsymbol{x}]^m$. We will often abuse notations to use $\boldsymbol{x}$ to also denote assignments over $\mathbb{F}_p^n$. For instance, let $\mathcal{V}_{\boldsymbol{f}} := \left\{\boldsymbol{x} \in \mathbb{F}_p^n : f_i(\boldsymbol{x}) = 0 \text{ for all } i \in [m]\right\}$ be the set of all common roots of $\boldsymbol{f}$.

Chevalley-Warning Theorem ([15, 37])**.** *For any prime*[1] *p and polynomial system $\boldsymbol{f} \in \mathbb{F}_p[\boldsymbol{x}]^m$ satisfying*

$$\sum_{i=1}^{m} \deg(f_i) < n, \qquad \text{(CW Condition)}$$

it holds that $|\mathcal{V}_{\boldsymbol{f}}| \equiv 0 \pmod{p}$.

Given a polynomial system $\boldsymbol{f} \in \mathbb{F}_p[\boldsymbol{x}]^m$, the key idea in the proof of the Chevalley-Warning Theorem is the polynomial

$$\mathsf{CW}_{\boldsymbol{f}}(\boldsymbol{x}) := \prod_{i=1}^{m} \left(1 - f_i(\boldsymbol{x})^{p-1}\right) \pmod{\left\{x_i^p - x_i\right\}_i} .$$

Observe that $\mathsf{CW}_{\boldsymbol{f}}(\boldsymbol{x}) = 1$ if $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{f}}$ and is 0 otherwise. Thus, $|\mathcal{V}_{\boldsymbol{f}}| \equiv \sum_{\boldsymbol{x} \in \mathbb{F}_p^n} \mathsf{CW}_{\boldsymbol{f}}(\boldsymbol{x}) \pmod{p}$. The following definition informally describes a special type of monomial of $\mathsf{CW}_{\boldsymbol{f}}$ that is of particular interest in the proof. For the precise definition, we refer to section 4.

Definition 2 (Max-Degree Monic Monomials (Informal))**.** Let $\boldsymbol{f} \in \mathbb{F}_p[\boldsymbol{x}]^m$. A *monic monomial* of $\mathsf{CW}_{\boldsymbol{f}}$ refers to a monic monomial obtained when symbolically expanding $\mathsf{CW}_{\boldsymbol{f}}$ as a sum of monic monomials. A monic monomial is said to be of *max-degree* if it is $\prod_{j=1}^{n} x_j^{p-1}$.

In the above definition, it is important to consider the *symbolic expansion* of $\mathsf{CW}_{\boldsymbol{f}}$ and ignore any cancellation of coefficients that might occur. Observe that, although the expansion of $\mathsf{CW}_{\boldsymbol{f}}$ is exponentially large in the description size of $\boldsymbol{f}$, each monic monomial of $\mathsf{CW}_{\boldsymbol{f}}$ can be succinctly described as a combination of monic monomials of the polynomials $f_1, \ldots, f_m$. We formally discuss this in section 4.

Using the definition of max-degree monic monomials, we state the main technical lemma underlying the proof of CWT (with proof in section 4).

[1] While most of the results in this section generalize to prime powers, we only consider prime fields for simplicity.

Chevalley–Warning Lemma. *For any prime p and $\boldsymbol{f} \in \mathbb{F}_p[\boldsymbol{x}]^m$,*

$$|\mathcal{V}_{\boldsymbol{f}}| \equiv (-1)^n \cdot |\{\text{max-degree monic monomials of } \mathsf{CW}_{\boldsymbol{f}}\}| \pmod p \quad \text{(CW Lemma)}$$

The Chevalley-Warning Theorem now follows by observing that if $\sum_{i=1}^m \deg(f_i) < n$ then the number of max-degree monic monomials of $\mathsf{CW}_{\boldsymbol{f}}$ is zero. Hence, we get that $|\mathcal{V}_{\boldsymbol{f}}| \equiv 0 \pmod p$.

1.2.2 Proofs of Cancellation

From the proof sketch of CWT in the previous section, a slight generalization of CWT follows. In particular, $|\mathcal{V}_{\boldsymbol{f}}| \equiv 0 \pmod p$ if and only if

$$\big|\{\text{max-degree monic monomials of } \mathsf{CW}_{\boldsymbol{f}}\}\big| \equiv 0 \pmod p, \quad \text{(Extended CW Condition)}$$

Thus, any condition on $\boldsymbol{f}$ that implies the (Extended CW Condition) can replace (CW Condition) in the Chevalley-Warning Theorem. Note that the (Extended CW Condition) is equivalent to all the max-degree monic monomials in $\mathsf{CW}_{\boldsymbol{f}}$ cancelling out. Thus, we call any such condition on $\boldsymbol{f}$ that implies (Extended CW Condition) to be a "*proof of cancellation*" for the system $\boldsymbol{f}$.

We can now reinterpret the result of Belovs et al. [8] in this framework of "proof of cancellation" conditions. In particular, [8] considers the case $p = 2$ and defines the problem PPA-Circuit-Chevalley, in which a "proof of cancellation" is given in a specific form of circuits. These circuits describe the system $(f_1, \ldots, f_m)$ in the PPA-Circuit-Chevalley problem. It is then shown that PPA-Circuit-Chevalley is PPA_2-complete.

1.2.3 Computational Problems Based on Chevalley-Warning Theorem

Every "proof of cancellation" that is *syntactically refutable* can be used to define a total search problem that lies in PPA_p. By *syntactically refutable* we mean that whenever the "proof of cancellation" is false, there exists a small witness that certifies so. In this section, we define three computational problems with their corresponding "proof of cancellation": (1) the Chevalley$_p$ problem defined by [33], (2) the GeneralChevalley$_p$ problem that is a generalization of Chevalley$_p$, and (3) the problem ChevalleyWithSymmetry$_p$ that we show to be PPA_p-complete. All these problems are defined for every prime modulus p and are natural in the sense that they do not explicitly involve a circuit or a Turing Machine in their input. In particular, the polynomial systems in the input are *explicit* in that they are given as a sum of monic monomials.

Chevalley. This is the direct computational analog of the Chevalley-Warning Theorem and was defined by Papadimitriou [33] as the following total search problem:

Chevalley$_p$
Given an explicit polynomial system $\boldsymbol{f} \in \mathbb{F}_p[\boldsymbol{x}]^m$, and an $\boldsymbol{x}^\star \in \mathcal{V}_{\boldsymbol{f}}$, output one of the following:

▷ [*Refuting witness*] (CW Condition) is not satisfied.
▷ $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{f}} \setminus \{\boldsymbol{x}^\star\}$.

We will particularly consider a special case where all the f_i's have zero constant term (*zecote*, for short). In this case, $\boldsymbol{x}^\star = \mathbf{0} \in \mathcal{V}_{\boldsymbol{f}}$, so there is no need to explicitly include $\boldsymbol{x}^*$ in the input.

General Chevalley. As mentioned already, we can define a search problem corresponding to any syntactically refutable condition that implies the (Extended CW Condition). One such condition is to directly assert that

$$\{\text{max-degree monic monomials of } \mathsf{CW}_{\boldsymbol{f}}\} = \emptyset. \quad \text{(General CW Condition)}$$

In particular, note that (CW Condition) implies this condition. Moreover, this condition is syntactically refutable by a max-degree monic monomial, which is efficiently representable as a combination of at most $m(p-1)$ monomials of the f_i's. Thus, we can define the following total search problem generalizing $\textsc{Chevalley}_p$.

$\underline{\textsc{GeneralChevalley}}_p$
Given an explicit polynomial system $\boldsymbol{f} \in \mathbb{F}_p[\boldsymbol{x}]^m$ and an $\boldsymbol{x}^\star \in \mathcal{V}_{\boldsymbol{f}}$, output one of the following:

▷ [*Refuting Witness*] A max-degree monic monomial of $\mathsf{CW}_{\boldsymbol{f}}$.
▷ $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{f}} \setminus \{\boldsymbol{x}^\star\}$.

While $\textsc{GeneralChevalley}_p$ generalizes $\textsc{Chevalley}_p$, it does not capture the full generality of (Extended CW Condition). However (Extended CW Condition) is not syntactically refutable (in fact, it is $\mathsf{Mod}_p\mathsf{P}$–complete to decide[2] if the final coefficient of the max-degree monomial is 0).

A natural question then is whether $\textsc{GeneralChevalley}_p$, or even $\textsc{Chevalley}_p$, could already be PPA_p–complete. We believe this to be unlikely because (General CW Condition) seems to fail in capturing other simple conditions that are syntactically refutable and yet imply (Extended CW Condition). Namely, consider a permutation $\sigma \in S_n$ of the variables $x_1, \ldots, x_n$ of order p (i.e. σ^p is the identity permutation). Suppose that for every $\boldsymbol{x} \in \overline{\mathcal{V}_{\boldsymbol{f}}}$, it holds that $\sigma(\boldsymbol{x}) \in \overline{\mathcal{V}_{\boldsymbol{f}}} \setminus \{\boldsymbol{x}\}$; in other words $\boldsymbol{x}, \sigma(\boldsymbol{x}), \sigma^2(\boldsymbol{x}), \ldots, \sigma^{p-1}(\boldsymbol{x})$ are all distinct and in $\overline{\mathcal{V}_{\boldsymbol{f}}}$ (where, $\sigma(\boldsymbol{x})$ denotes the assignment obtained by permutating the variables of the assignment $\boldsymbol{x}$ according to σ); observe that this condition is syntactically refutable. This implies that the elements of $\overline{\mathcal{V}_{\boldsymbol{f}}}$ can be partitioned into groups of size p (given by the orbits of the action σ) and hence $|\overline{\mathcal{V}_{\boldsymbol{f}}}| \equiv 0 \pmod p$. Thus, such a σ provides a syntactically refutable proof that $|\mathcal{V}_{\boldsymbol{f}}| \equiv 0 \pmod p$ and hence that (Extended CW Condition) hold.

Hence, we further generalize $\textsc{GeneralChevalley}_p$ into a problem that incorporates this additional "proof of cancellation" in the form of a permutation $\sigma \in S_n$.

[2] Circuit-SAT can be encoded as satisfiability of a polynomial system $\boldsymbol{f} \in \mathbb{F}_p[\boldsymbol{x}]^m$ by including a polynomial for each gate along with $\{x_i^2 - x_i = 0\}$ to ensure Booleanity. Thus, number of satisfiable assignments to the Circuit-SAT is $\equiv |\mathcal{V}_{\boldsymbol{f}}| \pmod p$, which is $0 \pmod p$ iff the final coefficient of the max-degree monomial is 0.

Chevalley with Symmetry. We consider a union of two polynomial systems $\boldsymbol{g} \in \mathbb{F}_p[\boldsymbol{x}]^{m_g}$ and $\boldsymbol{h} \in \mathbb{F}_p[\boldsymbol{x}]^{m_h}$. Even if both $\boldsymbol{g}$ and $\boldsymbol{h}$ satisfy (CW Condition), the combined system $\boldsymbol{f} := (g_1, \ldots, g_{m_g}, h_1, \ldots, h_{m_h})$ might not satisfy (CW Condition) and it might even be the case that $|\mathcal{V}_{\boldsymbol{f}}|$ is not a multiple of p. Thus, we need to bring in some additional conditions.

We start by observing that since $|\mathcal{V}_{\boldsymbol{f}}| + |\overline{\mathcal{V}_{\boldsymbol{f}}}| = p^n$, it holds that $|\mathcal{V}_{\boldsymbol{f}}| \equiv 0 \pmod{p}$ if and only if $|\overline{\mathcal{V}_{\boldsymbol{f}}}| \equiv 0 \pmod{p}$. Also note that, $|\overline{\mathcal{V}_{\boldsymbol{f}}}| = |\overline{\mathcal{V}_{\boldsymbol{g}}}| + |(\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}})|$.

If $\boldsymbol{g}$ satisfies the (General CW Condition) then we have that $|\mathcal{V}_{\boldsymbol{g}}| \equiv |\overline{\mathcal{V}_{\boldsymbol{g}}}| \equiv 0 \pmod{p}$. A simple way to enforce that $|\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}| \equiv 0 \pmod{p}$ is to enforce a "symmetry", namely that its elements can be grouped into groups of size p each. We impose this grouping with a permutation $\sigma \in S_n$ of the variables $x_1, \ldots, x_n$ of order p such that for any $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$, it holds that $\sigma(\boldsymbol{x}) \in (\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}) \setminus \{\boldsymbol{x}\}$; or in other words that $\boldsymbol{x}, \sigma(\boldsymbol{x}), \sigma^2(\boldsymbol{x}), \ldots, \sigma^{p-1}(\boldsymbol{x})$ are all distinct and contained in $\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$.

We now define the following natural total search problem.

ChevalleyWithSymmetry$_p$
Given two explicit polynomial systems $\boldsymbol{g} \in \mathbb{F}_p[\boldsymbol{x}]^{m_g}$ and $\boldsymbol{h} \in \mathbb{F}_p[\boldsymbol{x}]^{m_h}$, and an $\boldsymbol{x}^\star \in \mathcal{V}_{\boldsymbol{f}}$ (where $\boldsymbol{f} := (\boldsymbol{g}, \boldsymbol{h})$) and a permutation $\sigma \in S_n$ of order p, output one of the following:

- ▷ [*Refuting Witness* – 1] A max-degree monic monomial of $\mathsf{CW}_{\boldsymbol{g}}$.
- ▷ [*Refuting Witness* – 2] $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$ such that $\sigma(\boldsymbol{x}) \notin (\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}) \setminus \{\boldsymbol{x}\}$.
- ▷ $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{f}} \setminus \{\boldsymbol{x}^\star\}$.

The above problem is natural, because the input consists of a system of polynomial in an explicit form, i.e. as a sum of monic monomials, together with a permutation in S_n given say in one-line notation. Also, observe that when $\boldsymbol{h}$ is empty, the above problem coincides with GeneralChevalley$_p$ (since $\overline{\mathcal{V}_{\boldsymbol{h}}} = \emptyset$ when $\boldsymbol{h}$ is empty). Our main result is the following (proved in section 4).

Theorem 3. *For any prime p, ChevalleyWithSymmetry$_p$ is PPA_p-complete.*

2 The class $\mathbf{PPA}_q$

Search Problems in FNP and TFNP. A search problem in FNP is defined by a polynomial time computable relation $\mathcal{R} \subseteq \{0,1\}^* \times \{0,1\}^*$, that is, for every (x, y), it is possible to decide whether $(x, y) \in \mathcal{R}$ in $\mathrm{poly}(|x|, |y|)$ time. A solution to the search problem on input x is a y such that $|y| = \mathrm{poly}(|x|)$ and $(x, y) \in \mathcal{R}$. For convenience, define $\mathcal{R}(x) := \{y : (x, y) \in \mathcal{R}\}$. A search problem is *total* if for every input $x \in \{0,1\}^*$, there exists $y \in \mathcal{R}(x)$ such that $|y| \leq \mathrm{poly}(|x|)$. TFNP is the class of all total search problems in FNP.

Reducibility among search problems. A search problem $\mathcal{R}_1$ is *Karp-reducible* (or *many-one reducible*) to a search problem $\mathcal{R}_2$, or $\mathcal{R}_1 \preceq \mathcal{R}_2$ for short, if there exist polynomial-time computable functions f and g such that given any instance x of $\mathcal{R}_1$, $f(x)$ is an instance of $\mathcal{R}_2$ such that for any $y \in \mathcal{R}_2(f(x))$, it holds that $g(x, f(x), y) \in \mathcal{R}_1(x)$.

On the other hand, we say that $\mathcal{R}_1$ is *Turing-reducible* to $\mathcal{R}_2$, or $\mathcal{R}_1 \preceq_T \mathcal{R}_2$ for short, if there exists a polynomial-time oracle Turing machine that on input x to $\mathcal{R}_1$, makes oracle queries to $\mathcal{R}_2$, and outputs a $y \in \mathcal{R}_1(x)$. In this paper, we primarly deal with Karp-reductions, except in Section7.3, where we compare the two different notions of reductions in the context of PPA_q.

PPA_q via complete problems. We describe several total search problems (parameterized by q) that we show to be inter-reducible. PPA_q is then defined as the set of all search problems reducible to either one of the search problems defined below.

Recall that Boolean circuits take inputs of the form $\{0,1\}^n$ and operate using ($\wedge$, $\vee$, $\neg$) gates. In addition, we'll also consider circuits acting on inputs in $[q]^n$. We interpret the input to be of the form $(\{0,1\}^{\lceil \log q \rceil})^n$, where the circuit will be evaluated only on inputs where each block of $\lceil \log q \rceil$ bits represents a element in $[q]$. In the case where q is a prime, we could also represent the circuit as $C : \mathbb{F}_q^n \to \mathbb{F}_q^n$ with arbitrary gates of the form $g : \mathbb{F}_q^2 \to \mathbb{F}_q$. However, we can simulate any such gate with poly(q) many $+$ and $\times$ operations (over $\mathbb{F}_q$) along with a constant (1) gate. Hence, in the case of prime q, we'll assume that such circuits are composed of only $(+, \times, 1)$ gates.

Definition 6. (BIPARTITE$_q$)

Principle: A bipartite graph with a non-multiple-of-q degree node has another such node.

Object: Bipartite graph $G = (V \cup U, E)$. Designated vertex $v^* \in V$

Inputs: ▷ $C : \{0,1\}^n \to (\{0,1\}^n)^k$, with $(\{0,1\}^n)^k$ interpreted as a k-subset of $\{0,1\}^n$
▷ $v^* \in \{0\} \times \{0,1\}^{n-1}$ (usually 0^n)

Encoding: $V \coloneqq \{0\} \times \{0,1\}^{n-1}$, $U \coloneqq \{1\} \times \{0,1\}^{n-1}$,
$E \coloneqq \{(v,u) : v \in V \cap C(u) \text{ and } u \in U \cap C(v)\}$

Solutions: v^* if $\deg(v^*) \equiv 0 \pmod q$ and
$v \neq v^*$ if $\deg(v) \not\equiv 0 \pmod q$

Definition 7. (LONELY$_q$)

Principle: A q-dimensional matching on a non-multiple-of-q many vertices has an isolated node.

Object: q-dimensional matching $G = (V, E)$. Designated vertices $V^* \subseteq V$ with $|V^*| \leq q - 1$

Inputs: ▷ $C : [q]^n \to [q]^n$
▷ $V^* \subseteq [q]^n$ with $|V^*| \leq q - 1$

Encoding: $V \coloneqq [q]^n$. For distinct $v_1, \ldots, v_q$, edge $e \coloneqq \{v_1, \ldots, v_q\} \in E$ if $C(v_i) = v_{i+1}$, $C(v_q) = v_1$

Solutions: $v \in V^*$ if $\deg(v) = 1$ and
$v \notin V^*$ if $\deg(v) = 0$

Definition 8. ($\textsc{Leaf}_q$)

Principle: A q-uniform hypergraph with a non-multiple-of-q degree node has another such node.

Object: q-uniform hypergraph $G = (V, E)$. Designated vertex $v^* \in V$

Inputs: $\triangleright\ C : \{0,1\}^n \to (\{0,1\}^{nq})^q$, with $(\{0,1\}^{nq})^q$ interpreted as q many q-subsets of $\{0,1\}^n$
$\triangleright\ v^* \in \{0,1\}^n$ (usually 0^n)

Encoding: $V \coloneqq \{0,1\}^n$. For distinct $v_1, \ldots, v_q$, edge $e \coloneqq \{v_1, \ldots, v_q\} \in E$ if $e \in C(v)$ for all $v \in e$

Solutions: v^* if $\deg(v) \equiv 0 \pmod q$ and
$v \neq v^*$ if $\deg(v) \not\equiv 0 \pmod q$

We remark that $\textsc{Lonely}_q$ and $\textsc{Leaf}_q$ are modulo-q analogs of the PPA-complete problems Lonely and Leaf [33, 5]. We prove the following theorem in Appendix A.

Theorem 9. *The problems $\textsc{Bipartite}_q$, $\textsc{Lonely}_q$ and $\textsc{Leaf}_q$ are inter-reducible.*

Remark 10 (Simplifications in describing reductions.). We will use the following simple conventions repeatedly, in order to simplify the descriptions of reductions between different search problems.

1. We will often use "algorithms", instead of "circuits" to encode our hypergraphs. It is standard to simulate polynomial-time algorithms by polynomial sized circuits.
2. While our definitions require vertex sets to be of a very special form, e.g. $\{0,1\}^n$ or $[q]^n$, it will hugely simplify the description of our reductions to let vertex sets be of arbitrary sizes. This is not a problem as long as the vertex set is efficiently indexable, that is, elements of V must have a poly(n) length representation and we must have a poly-time computable bijective map $\varphi : V \to [|V|]$, whose inverse is also poly-time computable. We could then use φ to interpret the first $|V|$ elements of $\{0,1\}^n$ (or $[q]^n$) as vertices in V.
   Note that, we need to ensure that no new solutions are introduced in this process. In the case of $\textsc{Bipartite}_q$ or $\textsc{Leaf}_q$, we simply leave the additional vertices isolated and they don't contribute any new solutions. In the case of $\textsc{Lonely}_q$ we need to additionally ensure that $|V| \equiv 0 \pmod q$, so that we can easily partition the remaining vertices into q-uniform hyperedges thereby not introducing any new solutions.
3. The above simplification gives us that all our problems have an *instance-extension property* (cf. [10]) – this will be helpful in proving Theorem5.
4. To simplify our reductions even further, we'll often describe the edges/hyperdges directly instead of specifying how to compute the neighbors of a given vertex. This is only for simplicity and it will be easy to see how to compute the neighbors of any vertex locally.

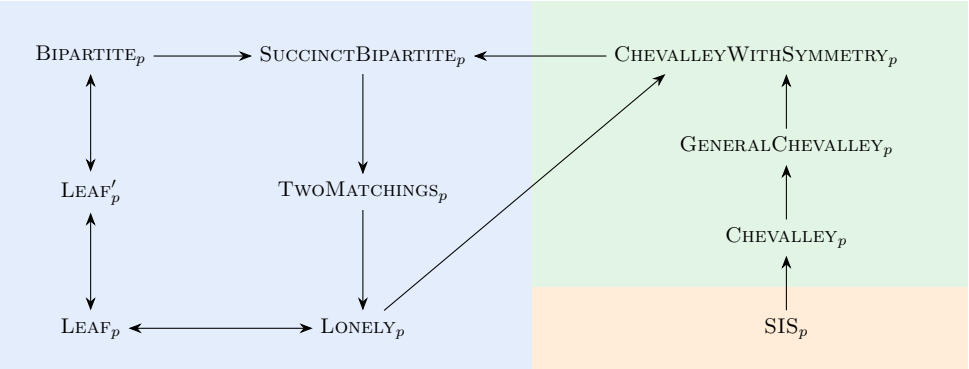


Figure 1: Total search problems studied in this work. An arrow $A \to B$ denotes a reduction $A \preceq B$ that we establish. Problems in the blue region are non-natural problems, which are all complete for PPA_p. Problems in the green region are natural problems of which ChevalleyWithSymmetry$_p$ is the one we show to be PPA_p–complete. The problem in the orange region is a cryptographically relevant problem.

4 A Natural Complete Problem

We start with some notation that will be useful for the presentation of our results.

Notations. For any polynomial $g \in \mathbb{F}_p[\boldsymbol{x}]$, we define $\deg(g)$ to be the degree of g. We define *the expansion to monic monomials* of g as $\sum_{\ell=1}^{L} t_\ell(\boldsymbol{x})$, where $t_\ell(\boldsymbol{x})$ is a monic monomial in $\mathbb{F}_p[\boldsymbol{x}]$, i.e. a monomial with coefficient 1. For example, the expansion of the polynomial $g(x_1, x_2) = x_1 \cdot (2x_1 + 3x_2)$ is given by $x_1^2 + x_1^2 + x_1x_2 + x_1x_2 + x_1x_2$.

For a polynomial system $\boldsymbol{f} := (f_1, \ldots, f_m) \in \mathbb{F}_p[\boldsymbol{x}]^m$, its affine variety $\mathcal{V}_{\boldsymbol{f}} \subseteq \mathbb{F}_p^n$ is defined as $\mathcal{V}_{\boldsymbol{f}} := \left\{\boldsymbol{x} \in \mathbb{F}_p^n \mid \boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{0}\right\}$. Let $\overline{\mathcal{V}_{\boldsymbol{f}}} := \mathbb{F}_p^n \setminus \mathcal{V}_{\boldsymbol{f}}$. If the constant term of each f_i is 0, we say that $\boldsymbol{f}$ is *zecote*, standing for "Zero Constant Term" (owing to lack of known terminology and creativity on our part).

4.1 The Chevalley-Warning Theorem

We repeat the formal statement of Chevalley-Warning Theorem together with its proof.

Chevalley-Warning Theorem ([15, 37])**.** *For any prime p and a polynomial system $\boldsymbol{f} \in \mathbb{F}_p[\boldsymbol{x}]^m$ satisfying $\sum_{i=1}^{m} \deg(f_i) < n$* (CW Condition)*, $|\mathcal{V}_{\boldsymbol{f}}| \equiv 0 \pmod{p}$.*

We describe the proof of CWT through Lemma 14. Even though there are direct proofs, the following presentation helps motivate the generalizations we study in future sections. Given a polynomial system $\boldsymbol{f} \in \mathbb{F}_p[\boldsymbol{x}]^m$, a key idea in the proof is the polynomial $\mathsf{CW}_{\boldsymbol{f}}(\boldsymbol{x}) := \prod_{i=1}^{m} \mathsf{CW}_{f_i}(\boldsymbol{x})$ where each $\mathsf{CW}_{f_i}(\boldsymbol{x}) := (1 - f_i(\boldsymbol{x})^{p-1})$. Observe that $\mathsf{CW}_{\boldsymbol{f}}(\boldsymbol{x}) = 1$ if $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{f}}$ and is 0 otherwise. The following definition describes the notion of a max-degree monomial of $\mathsf{CW}_{\boldsymbol{f}}$ that plays an important role in the proof.

Definition 13 (Max-Degree Monic Monomials)**.** For any prime p, let $\boldsymbol{f} \in \mathbb{F}_p[\boldsymbol{x}]^m$ and let the expansion into monic monomials of $\mathsf{CW}_{f_i}(\boldsymbol{x})$ be $\sum_{\ell=1}^{r_i} t_{i,\ell}(\boldsymbol{x})$. Let also $U_i = \{(i,\ell) \mid \ell \in [r_i]\}$ and $U = \times_{i=1}^m U_i$, we define the following quantities.

1. A *monic monomial* of $\mathsf{CW}_{\boldsymbol{f}}$ is a product $t_S(\boldsymbol{x}) = \prod_{i=1}^m t_{s_i}(\boldsymbol{x})$ for $S = (s_1, \ldots, s_m) \in U$.
2. A *max-degree monic monomial* of $\mathsf{CW}_{\boldsymbol{f}}$ is any monic monomial $t_S(\boldsymbol{x})$, such that $t_S(\boldsymbol{x}) \equiv \prod_{j=1}^n x_j^{p-1} \pmod{\{x_i^p - x_i\}_{i\in[n]}}$.
3. We define $\mathcal{M}_{\boldsymbol{f}}$ to be the set of max-degree monic monomials of $\mathsf{CW}_{\boldsymbol{f}}$, i.e. $\mathcal{M}_{\boldsymbol{f}} \coloneqq \{S \in U \mid t_S \text{ is a max-degree monic monomial of } \mathsf{CW}_{\boldsymbol{f}}\}$.

In words, the monomials $t(S)$ are precisely the ones that arise when symbolically expanding $\mathsf{CW}_{\boldsymbol{f}}(\boldsymbol{x})$. We illustrate this with an example: Let $p = 3$ and $f_1(x_1, x_2) = x_1 + x_2$ and $f_2(x_1, x_2) = x_1^2$. Then modulo $\{x_1^3 - x_1, x_2^3 - x_2\}$, we have

$$\begin{aligned}
\mathsf{CW}_{(f_1,f_2)}(x_1, x_2) &= (1 - (x_1 + x_2)^2)(1 - (x_1^2)^2) \\
&= (1 - x_1^2 - 2x_1x_2 - x_2^2) \cdot (1 - x_1^2) \\
&= (1 + x_1^2 + x_1^2 + x_1x_2 + x_2^2 + x_2^2) \cdot (1 + x_1^2 + x_1^2)
\end{aligned}$$

Thus there are 18 ($= 6 \times 3$) monic monomials in the system (f_1, f_2). The monomial corresponding to $S = ((1,5),(2,2))$ is a maximal monomial since the 5-th term in CW_{f_1} is x_2^2 and 2-nd term in CW_{f_2} is x_1^2. Using the above definitions, we now state the main technical lemma of the proof of CWT.

Lemma 14 (Main Lemma in the proof of CWT)**.** *For any prime p and any system of polynomials $\boldsymbol{f} \in \mathbb{F}_p[\boldsymbol{x}]^m$, it holds that $|\mathcal{V}_{\boldsymbol{f}}| \equiv (-1)^n |\mathcal{M}_{\boldsymbol{f}}| \pmod p$.*

Proof. As noted earlier, $\mathsf{CW}_{\boldsymbol{f}}(\boldsymbol{x}) = 1$ if $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{f}}$ and is 0 otherwise. Thus, it follows that $|\mathcal{V}_{\boldsymbol{f}}| \equiv \sum_{\boldsymbol{x}\in\mathbb{F}_p^n} \mathsf{CW}_{\boldsymbol{f}}(\boldsymbol{x}) \pmod p$. For any monic monomial $m(\boldsymbol{x}) = \prod_{j=1}^n x_j^{d_j}$, it holds that $\sum_{\boldsymbol{x}\in\mathbb{F}_p^n} m(\boldsymbol{x}) = 0$ if $d_j < p-1$ for some x_j. On the other hand, for the monic max-degree monomial $m(\boldsymbol{x}) = \prod_{j=1}^n x_j^{p-1}$, it holds that $\sum_{\boldsymbol{x}\in\mathbb{F}_p^n} m(\boldsymbol{x}) = (p-1)^n$. Thus, we get that $|\mathcal{V}_{\boldsymbol{f}}| \equiv \sum_{\boldsymbol{x}\in\mathbb{F}_p^n} \mathsf{CW}_{\boldsymbol{f}}(\boldsymbol{x}) \pmod p \equiv \sum_{S\in U} \sum_{\boldsymbol{x}\in\mathbb{F}_p^n} t_S(\boldsymbol{x}) \pmod p \equiv (-1)^n |\mathcal{M}_{\boldsymbol{f}}| \pmod p$. □

The proof of Chevalley-Warning Theorem follows easily from Lemma 14.

Proof of Chevalley-Warning Theorem. We have that $\deg(\mathsf{CW}_{\boldsymbol{f}}) \le (p-1)\sum_{i=1}^m \deg(f_i)$. Thus, if $\boldsymbol{f}$ satisfies (CW Condition), then $\deg(\mathsf{CW}_{\boldsymbol{f}}) < (p-1)n$ and hence $|\mathcal{M}_{\boldsymbol{f}}| = 0$. CWT now follows from Lemma 14. □

4.2 The Chevalley-Warning Theorem with Symmetry

In this section, we formalize the intuition that we built in Sections 1.2.2 and 1.2.3 to prove the more general statements to lead to the same conclusion as the Chevalley-Warning Theorem.

First, we prove a theorem that argues about the cardinality of $\mathcal{V}_{\boldsymbol{f}}$ directly using some symmetry of the system of polynomials $\boldsymbol{f}$. Then, combining this

symmetry-based argument with the (General CW Condition) we get the generalization of the Chevalley-Warning Theorem. Our natural PPA_p-complete problem is based on this generalization.

The theorem statements are simplified using the definition of *free action* of a group. For a permutation over n elements $\sigma \in S_n$, we define $\langle\sigma\rangle$ to be the sub-group generated by σ and $|\sigma|$ to be the order of $\langle\sigma\rangle$. For $\boldsymbol{x} \in \mathbb{F}_p^n$, $\sigma(\boldsymbol{x})$ denotes the assignment obtained by permutating the variables of the assignment $\boldsymbol{x}$ according to σ.

Definition 15 (Free Group Action)**.** Let $\sigma \in S_n$ and $\mathcal{V} \subseteq \mathbb{F}_p^n$, then we say that $\langle\sigma\rangle$ acts freely on $\mathcal{V}$ if, for every $\boldsymbol{x} \in \mathcal{V}$, it holds that $\sigma(\boldsymbol{x}) \in \mathcal{V}$ and $\boldsymbol{x} \neq \sigma(\boldsymbol{x})$.

Our first theorem highlights the use of symmetry in arguing about the size of $|\mathcal{V}_{\boldsymbol{f}}|$.

Theorem 16. *Let $\boldsymbol{f} \in \mathbb{F}_p[\boldsymbol{x}]^m$ be a system of polynomials. If there exists a permutation $\sigma \in S_n$ with $|\sigma| = p$ such that $\langle\sigma\rangle$ acts freely on $\overline{\mathcal{V}_{\boldsymbol{f}}}$, then $|\mathcal{V}_{\boldsymbol{f}}| \equiv 0 \pmod p$.*

Proof. Since σ acts freely on $\overline{\mathcal{V}_{\boldsymbol{f}}}$, we can partition $\overline{\mathcal{V}_{\boldsymbol{f}}}$ into orbits of any $\boldsymbol{x} \in \overline{\mathcal{V}_{\boldsymbol{f}}}$ under actions of $\langle\sigma\rangle$, namely sets of the type $\left\{\sigma^i(\boldsymbol{x})\right\}_{i\in[p]}$ for $\boldsymbol{x} \in \overline{\mathcal{V}_{\boldsymbol{f}}}$. Since $\langle\sigma\rangle$ acts freely on $\overline{\mathcal{V}_{\boldsymbol{f}}}$, each such orbit has size p. Thus, we can conclude that $\left|\overline{\mathcal{V}_{\boldsymbol{f}}}\right| \equiv 0 \pmod p$ from which the theorem follows. □

Remark 17. For any polynomial system $\boldsymbol{f}$ and any permutation σ, we can check in linear time if $|\sigma| = p$ and we can syntactically refute that $\langle\sigma\rangle$ acts freely on $\overline{\mathcal{V}_{\boldsymbol{f}}}$ with an $\boldsymbol{x} \in \mathbb{F}_p^n \setminus \{\boldsymbol{0}\}$ such that $\boldsymbol{f}(\sigma(\boldsymbol{x})) = \boldsymbol{0}$ or $\sigma(\boldsymbol{x}) = \boldsymbol{x}$.

We now state and prove an extension of CWT that captures both the argument from Lemma 14 and the symmetry argument from Theorem 16.

Theorem 18 (Chevalley-Warning with Symmetry Theorem)**.** *Let $\boldsymbol{g} \in \mathbb{F}_p[\boldsymbol{x}]^{m_g}$ and $\boldsymbol{h} \in \mathbb{F}_p[\boldsymbol{x}]^{m_h}$ be two systems of polynomials, and $\boldsymbol{f} \coloneqq (\boldsymbol{g}, \boldsymbol{h})$. If there exists a permutation $\sigma \in S_n$ with $|\sigma| = p$ such that (1) $\mathcal{M}_{\boldsymbol{g}} = \emptyset$ and (2) $\langle\sigma\rangle$ acts freely on $\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$, then $|\mathcal{V}_{\boldsymbol{f}}| = 0 \pmod p$.*

Remark 19. We point to the special form of Condition 2. By definition, $\mathcal{V}_{\boldsymbol{f}} = \mathcal{V}_{\boldsymbol{g}} \cap \mathcal{V}_{\boldsymbol{h}}$, hence if $\langle\sigma\rangle$ were to act freely on $\overline{\mathcal{V}_{\boldsymbol{g}}} \cup \overline{\mathcal{V}_{\boldsymbol{h}}}$ (or even $\mathcal{V}_{\boldsymbol{g}} \cap \mathcal{V}_{\boldsymbol{h}}$), then we could just use Theorem 16 to get that $|\mathcal{V}_{\boldsymbol{f}}| \equiv 0 \pmod p$. In the above theorem, we only require that $\langle\sigma\rangle$ acts freely on $\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$. Observe that Theorem 16 follows as a special case of CWT with Symmetry by setting $m_g = 0$. Additionally, by setting $m_h = 0$ we get the generalization of CWT corresponding to the (General CW Condition) as presented in subsubsection 1.2.3.

Proof of Theorem 18. If $\mathsf{CW}_{\boldsymbol{g}}$ does not have any max- degree monic monomials, we have $|\mathcal{V}_{\boldsymbol{g}}| \equiv 0 \pmod p$ (similar to proof of CWT) and, since $\overline{\mathcal{V}_{\boldsymbol{g}}} = \mathbb{F}_p^n \setminus \mathcal{V}_{\boldsymbol{g}}$, we have $\left|\overline{\mathcal{V}_{\boldsymbol{g}}}\right| \equiv 0 \pmod p$. Also, since $\langle\sigma\rangle$ acts freely on $\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$, we have $\left|\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}\right| \equiv 0 \pmod p$ (similar to the proof of Theorem 16). Hence, $\left|\overline{\mathcal{V}_{\boldsymbol{f}}}\right| = \left|\overline{\mathcal{V}_{\boldsymbol{g}} \cap \mathcal{V}_{\boldsymbol{h}}}\right| = \left|\overline{\mathcal{V}_{\boldsymbol{g}}} \cup \overline{\mathcal{V}_{\boldsymbol{h}}}\right| = \left|\overline{\mathcal{V}_{\boldsymbol{g}}}\right| + \left|\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}\right| \equiv 0 \pmod p$. Thus, $|\mathcal{V}_{\boldsymbol{f}}| \equiv 0 \pmod p$. □

4.3 Computational Problems Related to Chevalley-Warning Theorem

We now follow the intuition developed in the previous section and in subsection 1.2 to formally define the computational problems CHEVALLEY_p, $\text{GENERALCHEVALLEY}_p$, and $\text{CHEVALLEYWITHSYMMETRY}_p$.

Definition 20. (CHEVALLEY_p)

Principle: Chevalley-Warning Theorem.

Input: $\boldsymbol{f} \in \mathbb{F}_p[\boldsymbol{x}]^m$: an explicit zecote polynomial system.

Condition: $\sum_{i=1}^{m} \deg(f_i) < n$.

Output: $\boldsymbol{x} \in \mathbb{F}_p^n$ such that $\boldsymbol{x} \neq \boldsymbol{0}$ and $\boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{0}$.

Definition 21. ($\text{GENERALCHEVALLEY}_p$)

Principle: General Chevalley-Warning Theorem via (General CW Condition).

Input: $\boldsymbol{f} \in \mathbb{F}_p[\boldsymbol{x}]^m$: an explicit zecote polynomial system.

Output: 0. A max-degree monic monomial $t_S(\boldsymbol{x})$ of $\mathsf{CW}_{\boldsymbol{f}}$, or
1. $\boldsymbol{x} \in \mathbb{F}_p^n$ such that $\boldsymbol{x} \neq \boldsymbol{0}$ and $\boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{0}$.

Definition 22. ($\text{CHEVALLEYWITHSYMMETRY}_p$)

Principle: Chevalley-Warning Theorem with Symmetry (Theorem 18).

Input: ▷ $\boldsymbol{g} \in \mathbb{F}_p[\boldsymbol{x}]^{m_g}$ and $\boldsymbol{h} \in \mathbb{F}_p[\boldsymbol{x}]^{m_h}$: explicit zecote polynomial systems
▷ $\sigma \in S_n$: a permutation over $[n]$.

Condition: $|\sigma| = p$.

Output: 0. (a) A max-degree monic monomial $t_S(\boldsymbol{x})$ of $\mathsf{CW}_{\boldsymbol{g}}$, or
(b) $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$ such that $\sigma(\boldsymbol{x}) \notin (\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}) \setminus \{\boldsymbol{x}\}$, or
1. $\boldsymbol{x} \in \mathbb{F}_p^n$ such that $\boldsymbol{x} \neq \boldsymbol{0}$ and $\boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{0}$.

Remark 23. Some observations about the above computational problems follow:

1. In the problems $\text{GENERALCHEVALLEY}_p$ and $\text{CHEVALLEYWITHSYMMETRY}_p$, we assume that, if the output is a max-degree monic monomial, this is given via the multiset of indices S that describes the monomial as formalized in Definition 13.
2. We have $\text{CHEVALLEY}_p \preceq \text{GENERALCHEVALLEY}_p \preceq \text{CHEVALLEYWITHSYMMETRY}_p$. Thus, inclusion of $\text{CHEVALLEYWITHSYMMETRY}_p$ in PPA_p implies that the problems CHEVALLEY_p and $\text{GENERALCHEVALLEY}_p$ are in PPA_p. Also, in Section6 we prove that $\text{SIS}_p \preceq \text{CHEVALLEY}_p$, where SIS_p is a cryptographically relevant problem. This shows that the $\text{GENERALCHEVALLEY}_p$ and the $\text{CHEVALLEYWITHSYMMETRY}_p$ problems are at least as hard as the SIS_p problem.

We restate our main result.

Theorem 3. *For any prime p, $\text{CHEVALLEYWITHSYMMETRY}_p$ is PPA_p-complete.*

4.4 ChevalleyWithSymmetry$_p$ is PPA$_p$–complete

We fist prove that ChevalleyWithSymmetry is in PPA$_p$ and then prove its PPA$_p$-hardness.

4.4.1 ChevalleyWithSymmetry$_p$ is in PPA$_p$

Even though Papadimitriou [33] provided a rough proof sketch of Chevalley$_p \in$ PPA$_p$, a formal proof was not given. We show that ChevalleyWithSymmetry$_p$ is in PPA$_p$ (and so are GeneralChevalley$_p$ and Chevalley$_p$). In order to do so we extend the definition of Bipartite$_q$ to instances where the vertices might have exponential degree and edges appear with multiplicity. The key here is to define a Bipartite$_q$ instance with unbounded (even exponential) degree, but with additional information that allows us to verify solutions efficiently.

Definition 24. (SuccinctBipartite$_q$)

Principle: Similar to Bipartite$_q$, but degrees are allowed to be exponentially large, edges are allowed with multiplicities at most $q-1$.

Object: Bipartite graph $G = (V \cup U, E)$ s.t. $E \subseteq V \times U \times \mathbb{Z}_q$. Designated edge $e^* \in E$.

Inputs: Let $V \coloneqq \{0\} \times \{0,1\}^{n-1}$ and $U \coloneqq \{1\} \times \{0,1\}^{n-1}$:
▷ $\mathcal{C} : V \times U \to [q]$, edge counting circuit
▷ $\phi_V : V \times U \times [q] \to (U \times [q])^q$, grouping pivoted at V
▷ $\phi_U : V \times U \times [q] \to (V \times [q])^q$, grouping pivoted at U
▷ $e^* = (v^*, u^*, k^*)$, designated edge

Encoding: $V \coloneqq \{0\} \times \{0,1\}^{n-1}$, $U \coloneqq \{1\} \times \{0,1\}^{n-1}$,
$E \coloneqq \{(v,u,k) : 1 \le k \le C(v,u),\ (v,u) \in V \times U\}$ (here k distinguishes multiplicities)
Edge (v,u,k) is grouped with $\{(v,u',k') : (u',k') \in \phi_V(v,u,k)\}$ (pivoting at v),
provided $|\phi_V(v,u,k)| = q$, all $(v,u',k') \in E$ and $\phi_V(v,u',k') = \phi_V(v,u,k)$.
Edge (v,u,k) is grouped with $\{(v',u,k') : (v',k') \in \phi_U(v,u,k)\}$ (pivoting at u),
provided $|\phi_U(v,u,k)| = q$, all $(v,u',k') \in E$ and $\phi_U(v',u,k') = \phi_V(v,u,k)$.

Solutions: e^* if e^* is grouped, pivoting at v^*, or if e^* is not grouped pivoting at u^*, OR
$e \neq e^*$ if e is not grouped pivoting at one of its ends.

In words, SuccinctBipartite$_p$ encodes a bipartite graph with arbitrary degree. Instead of listing the neighbors of a vertex using a circuit, we have a circuit that outputs the multiplicity of edges between any two given vertices. We are therefore unable to efficiently count the number of edges incident on any vertex. The grouping function ϕ_V aims to group edges incident on any vertex $v \in V$ into groups of size q. Similarly, ϕ_U aims to group edges incident on any vertex $u \in U$. The underlying principle is that if we have an edge e^* that is not grouped pivoting at v^* (one of its endpoints), then either e^* is not pivoted at u^* (its other endpoint) or there exists another edge that is also not grouped

pivoting at one of its ends. Note that in contrast to the problems previously defined, v^* might still be an endpoint of a valid solution.

Lemma 25. *For all primes p, $\textsc{ChevalleyWithSymmetry}_p \in \mathsf{PPA}_p$.*

Proof. We reduce $\textsc{ChevalleyWithSymmetry}_p$ to $\textsc{SuccinctBipartite}_p$, which we show to be PPA_p–complete in subsection A.1. Given an instance of $\textsc{ChevalleyWithSymmetry}_p$, namely a zecote polynomial system $\boldsymbol{f} = (\boldsymbol{g}, \boldsymbol{h})$ and a permutation σ, we construct a bipartite graph $G = (U \cup V, E)$ encoded as an instance of $\textsc{SuccinctBipartite}_p$ as follows.

Description of vertices. $U = \mathbb{F}_p^n$, namely all possible assignments of $\boldsymbol{x}$. The vertices of V are divided into two parts $V_1 \cup V_2$. The part V_1 contains one vertex for each monomial in the expansion of $\mathsf{CW}_{\boldsymbol{g}} = \prod_{i=1}^{m_g}(1 - g_i^{p-1})$. Since p is constant, we can efficiently list out the monomials of $1 - g_i^{p-1}$. For a fixed lexicographic ordering of the monomials of each $\mathsf{CW}_{g_i} \coloneqq 1 - g_i^{p-1}$, a monomial of $\mathsf{CW}_{\boldsymbol{g}}$ is represented by a tuple $(a_1, a_2, \ldots, a_{m_g})$ with $0 \le a_i < L_i$, where a_i represents the index of a monomial of CW_{g_i} and L_i is the number of monomials of CW_{g_i}, where $a_i = 0$ corresponds to the constant term 1. The part $V_2 \coloneqq \binom{\mathbb{F}_p^n}{p}$, i.e. it contains a vertex for each subset of p distinct elements in $\mathbb{F}_p^n$.

Description of edges. We first describe the edges between U and V_1, namely include an edge between an assignment $\boldsymbol{x}$ and a monomial t with multiplicity $t(\boldsymbol{x})$. With these edges in place, the degree of vertices are as follows:

- $\boldsymbol{x} = 0^n$ has a single edge corresponding to the constant monomial 1, since $\boldsymbol{f}$ is zecote. We let this be the designated edge e^* in the final $\textsc{SuccinctBipartite}_p$ instance.
- $\boldsymbol{x} \notin \mathcal{V}_{\boldsymbol{g}}$ has $0 \pmod p$ edges (counting multiplicities). Since $\mathsf{CW}_{\boldsymbol{g}}(\boldsymbol{x}) = 0$, the sum over all monomials of $t(\boldsymbol{x})$ must be $0 \pmod p$.
- $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}}$ has $1 \pmod p$ edges (counting multiplicities), since the sum over all $t(\boldsymbol{x})$ monomials gives $\mathsf{CW}_{\boldsymbol{g}}(\boldsymbol{x}) \equiv 1 \pmod p$.

Thus with the edges so far, the vertices (excluding 0^n), with degree $\not\equiv 0 \pmod p$ are precisely vertices $t \in V_1$ such that $\sum_{\boldsymbol{x}} t(\boldsymbol{x}) \not\equiv 0 \pmod p$ or $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \setminus \{0^n\}$. For the former case, if t contained a variable with degree less than $p-1$, then $\sum_{\boldsymbol{x}} t(\boldsymbol{x}) \equiv 0 \pmod p$. Hence, it must be that $t = \prod_{i=1}^{n} x_i^{p-1}$. In the later case, the degree of $\boldsymbol{x}$ is $1 \pmod p$ and hence $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}}$. However, there is no guarantee that a vertex $\boldsymbol{x}$ with degree $1 \pmod p$ is in $\mathcal{V}_{\boldsymbol{h}}$ as well. To argue about $\boldsymbol{h}$, we add edges between U and V_2 that exclude solutions $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$, on which σ acts freely (that is, $\sigma(\boldsymbol{x}) = \boldsymbol{x}$). More specifically, for $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$, if $\sigma(\boldsymbol{x}) \neq \boldsymbol{x}$, we add an edge with multiplicity $p-1$ between $\boldsymbol{x}$ and $\Sigma_{\boldsymbol{x}} \in V_2$ where $\Sigma_{\boldsymbol{x}} \coloneqq \{\sigma^i(\boldsymbol{x})\}_{i \in \mathbb{Z}_p}$ (note that, in this case $|\Sigma_{\boldsymbol{x}}| = p$ since $\sigma(\boldsymbol{x}) \neq \boldsymbol{x}$ and $|\sigma| = p$ is prime). Observe that, if a vertex in V_2 corresponds to a $\Sigma_{\boldsymbol{x}}$, it has p edges each with multiplicity $p-1$, one for each $\boldsymbol{x}' \in \Sigma_{\boldsymbol{x}}$ only if $\Sigma_{\boldsymbol{x}} \subseteq \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$. If a vertex in V_2 does not correspond to a $\Sigma_{\boldsymbol{x}}$, then it has no edges. Thus, a vertex in V_2 has degree $\not\equiv 0 \pmod p$ iff it contains an $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$ such that $\sigma(\boldsymbol{x}) \notin \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$.

Thus, with all the edges added, vertices with degree $\not\equiv 0 \pmod p$ correspond to one of

- $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \mathcal{V}_{\boldsymbol{h}}$ such that $\boldsymbol{x} \neq \boldsymbol{0}$, or

- $t \in V_1$ such that $t(\boldsymbol{x})$ is a max-degree monomial or
- $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$ such that $\sigma(\boldsymbol{x}) = \boldsymbol{x}$ or
- $v \in V_2$ such that $\exists\, \boldsymbol{x} \in v$ satisfying $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$ and $\sigma(\boldsymbol{x}) \notin \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$.

These correspond precisely to the solutions of ChevalleyWithSymmetry$_p$. To summarize, the edge counting circuit C on input $(\boldsymbol{x}, t) \in U \times V_1$ outputs $t(\boldsymbol{x})$ and on input $(\boldsymbol{x}, v) \in U \times V_2$ outputs $p-1$ if $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$, $\sigma(\boldsymbol{x}) \neq \boldsymbol{x}$ and $v = \Sigma_{\boldsymbol{x}}$ and 0 otherwise.

Grouping Functions. The grouping functions ϕ_U and ϕ_V are defined as follows (analogous to the so-called "chessplayer algorithm" in [33]):

▷ Grouping ϕ_U (corresponding to endpoint in U):

- For $\boldsymbol{x} \in \overline{\mathcal{V}_{\boldsymbol{g}}}$: there exists some i such that $\mathsf{CW}_{g_i}(\boldsymbol{x}) = 0$. Consider an edge $(\boldsymbol{x}, (a_1, a_2, \ldots, a_{m_g}), k)$. We can explicitly list out the multiset containing the monomials $t_j = (a_1, a_2, \ldots, a_i \leftarrow j, \ldots, a_{m_g})$ with multiplicity $t_j(\boldsymbol{x})$, for each $1 \leq j \leq L_i$. Since $\mathsf{CW}_{g_i}(\boldsymbol{x}) = 0$, this multiset has size multiple of p. Hence, we can canonically divide its elements into groups of size p, counting multiplicities and ϕ_U returns the subset containing (t, k).
- For $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$ such that $\sigma(\boldsymbol{x}) \neq \boldsymbol{x}$: Note that $g_i^{p-1}(\boldsymbol{x}) = 0$ for all $i \in [m_g]$. Let $v_1 \in V_1$ be the vertex corresponding to the constant monomial 1. ϕ_U groups the edge $(\boldsymbol{x}, v_1, 1)$ (of multiplicity 1) with the $p-1$ edges $(\boldsymbol{x}, \Sigma_{\boldsymbol{x}}, k)$ for $k \in [p-1]$. For any other $t \in V_1 \setminus \{v_1\}$ and an edge $(\boldsymbol{x}, t, k)$, we have that $t = (a_1, \ldots, a_{m_g})$ has $a_i \neq 0$ for some i. We define the multiset containing $t_j = (a_1, \ldots, a_i \leftarrow j, \ldots a_{m_g})$ with multiplicity $t_j(\boldsymbol{x})$ for each $1 \leq j < L_i$. Since $g_i^{p-1}(\boldsymbol{x}) = 0$, this multiset has size which is a multiple of p, which we can canonically partition into groups of size p. Thus, ϕ_U on input $(\boldsymbol{x}, t, k)$ returns the group containing (t, k).

▷ Grouping ϕ_V (corresponding to endpoint in V):

- For $t \in V_1$ such that $t \neq \prod_{i=1}^{n} x_i^{p-1}$: there exists a variable x_i with degree less than $p-1$. For $\boldsymbol{x}_j = (x_1, \ldots, x_{i-1}, x_i \leftarrow j, \ldots, x_n)$ with $j \in \mathbb{F}_p$ we define the multiset $\{(\boldsymbol{x}_j, t(\boldsymbol{x}_j))\}_{j \in \mathbb{F}_p}$. Since $\sum_{j=0}^{p-1} t(\boldsymbol{x}_j) = 0$, this multiset has size multiple of p, so we can canonically partition it into groups of size p. Then, $\phi_V(\boldsymbol{x}, t, k)$ returns the group containing $(\boldsymbol{x}, k)$,
- For $v \in V_2$: if $\deg(v) = 0$, then there is no grouping to be done. Else if $\deg(v) \equiv 0 \pmod p$ then $\phi_V(\boldsymbol{x}, t, k)$ returns $\{(\boldsymbol{x}, k)\}_{\boldsymbol{x} \in v}$.

Thus, for any vertex with degree $\equiv 0 \pmod p$, we have provided a grouping function for all its edges. So, for any edge that is not grouped by grouping function at any of its endpoints, then such an endpoint must have degree $\not\equiv 0 \pmod p$ and hence point to a valid solution of the ChevalleyWithSymmetry$_p$ instance. □

4.4.2 ChevalleyWithSymmetry$_p$ is PPA$_p$–hard

We show that Lonely$_p \preceq$ ChevalleyWithSymmetry$_p$. In the ChevalleyWithSymmetry$_p$ instance that we create, we will ensure that there are no solutions of type 0 (as in Definition 22) and thus, the only valid solutions will be of type 1. In order

to do so, we introduce the notions of labeling and proper labeling and prove a generalization of CWT that we call Labeled CWT (Theorem 30).

As we will see, the Labeled CWT, is just a re-formulation of the original CWT rather than a generalization. To understand the Labeled CWT we start with some examples that do not seem to satisfy the Chevalley-Warning condition, but where a solution exists.

Example 1. Consider the case where $p = 3$ and $f(x_1, x_2) = x_2 - x_1^2$. In this case the Chevalley-Warning condition is not met, since we have 2 variables and the total degree is also 2. But, let us consider a slightly different polynomial where we replace the variable x_2 with the product of two variables x_{21}, x_{22} then we get the polynomial $g(x_1, x_{21}, x_{22}) = x_{21} \cdot x_{22} - x_1^2$. Now, g satisfies (CW Condition) and hence, we conclude that the number of roots of g is a multiple of 3. Interestingly, from this fact we can argue that there exists a non-trivial solution for $f(\boldsymbol{x}) = 0$. In particular, the assignment $x_1 = 0$, $x_2 = 0$ corresponds to five assignments of the variables x_1, x_{21}, x_{22}. Hence, since $|\mathcal{V}_g| = 0 \pmod 3$, g has another root, which corresponds to a non-trivial root of f. In this example, we applied the CWT on a slightly different polynomial than f to argue about the existence of non-trivial solutions of f, even though f did not satisfy (CW Condition) itself.

Ignore Some Terms. The Labeled CWT formalizes the phenomenon observed in Example 1 and shows that under certain conditions we can *ignore some terms* when defining the degree of each polynomial. For instance, in Example 1, we can ignore the term x_1^2 when computing the degree of f and treat f as a degree 1 polynomial of 2 variables, in which case the condition of CWT is satisfied.

We describe which terms can be ignored by defining a *labeling* of the terms of each polynomial in the system. The labels take values in $\{-1, 0, +1\}$ and our final goal is to ignore the terms with label $+1$. Of course, it should not be possible to define any labeling that we want; for example we cannot ignore all the terms of a polynomial. Next, we describe the rules of a *proper labeling* that will allow us to prove the Labeled CWT. We start with a definition of a labeling.

Definition 26 (Monomial Labeling)**.** Let $\boldsymbol{f} \in \mathbb{F}[\boldsymbol{x}]^m$ and let t_{ij} be the j-th monomial of the polynomial $f_i \in \mathbb{F}[\boldsymbol{x}]$ (written in some canonical sorted order). Let $\mathcal{T}$ be the set of all pairs (i, j) such that t_{ij} is a monomial in $\boldsymbol{f}$. A *labeling* of $\boldsymbol{f}$ is a function $\lambda : \mathcal{T} \to \{-1, 0, +1\}$ and we say that $\lambda(i, j)$ is the *label* of t_{ij} according to λ.

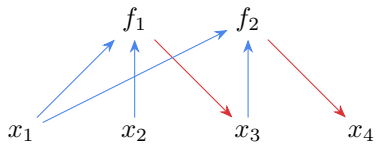
Definition 27 (Labeled Degree)**.** For $\boldsymbol{f} \in \mathbb{F}[\boldsymbol{x}]^m$ with a labeling λ, we define the *labeled degree* of f_i as, $\deg^{\lambda}(f_i) \coloneqq \max_{j\,:\,\lambda(i,j)\neq+1} \deg(t_{ij})$, in words, maximum degree among monomials of f_i labeled either 0 or -1.

Example 1 (continued). According to the lexicographic ordering, $f(x_1, x_2) = -x_1^2 + x_2$ and we have the monomials $t_{11} = -x_1^2$ and $t_{12} = x_2$. Hence, one possible labeling, which as we will see later corresponds to the vanilla Chevalley-Warning Theorem, is $\lambda(1, 1) = \lambda(1, 2) = 0$. According to this labeling, $\deg^{\lambda}(f) = 2$. Another possible labeling, that, as we will see, allows us to apply the Labeled CWT , is $\lambda(1, 1) = +1$ and $\lambda(1, 2) = -1$. In this case, the labeled degree is $\deg^{\lambda}(f) = 1$.

As we highlighted before, our goal is to prove the Chevalley-Warning Theorem, but with the weaker condition that $\sum_{i=1}^{m} \deg^{\lambda}(f_i) < n$ instead of $\sum_{i=1}^{m} \deg(f_i) < n$. Of course, we first have to restrict the space of all possible labelings by defining *proper labelings*. In order to make the condition of proper labelings easier to interpret we start by defining the notion of a labeling graph.

Definition 28 (Labeling Graph)**.** For $\boldsymbol{f} \in \mathbb{F}[\boldsymbol{x}]^m$ with a labeling λ, we define the *labeling graph* $G_\lambda = (U \cup V, E)$ as a directed bipartite graph on vertices $U = \{x_1, \ldots, x_n\}$ and $V = \{f_1, \ldots, f_m\}$. The edge $(x_j \to f_i)$ belongs to E if x_j appears in a monomial t_{ir} in f_i with label $+1$, i.e. $\lambda(i, r) = +1$. Symmetrically, the edge $(f_i \to x_j)$ belongs to E if the x_j appears in a monomial t_{ir} in f_i with label -1, i.e. $\lambda(i, r) = -1$.

Example 2. Let $p = 2$ and $f_1(x_1, x_2, x_3, x_4) = x_1x_2 - x_3$, $f_2(x_1, x_2, x_3, x_4) = x_1x_3 - x_4$. In this system, if we use the lexicographic monomial ordering we have the monomials $t_{11} = x_1x_2$, $t_{12} = -x_3$, $t_{21} = x_1x_3$, $t_{22} = -x_4$. The following figure shows the graph G_λ for the labeling $\lambda(1, 1) = +1$, $\lambda(1, 2) = -1$, $\lambda(2, 1) = +1$ and $\lambda(2, 2) = -1$.



Definition 29 (Proper Labeling)**.** Let $\boldsymbol{f} \in \mathbb{F}[\boldsymbol{x}]^m$ with a labeling λ. We say that the labeling λ is *proper* if the following conditions hold.

(1) For all i, either $\lambda(i, j) \in \{-1, 1\}$ for all j, or $\lambda(i, j) = 0$ for all j.
(2) If two monomials t_{ij}, $t_{ij'}$ contain the same variable x_k, then $\lambda(i, j) = \lambda(i, j')$.
(3) If $\lambda(i, j) = -1$, then t_{ij} is multilinear.
(4) If x_k is a variable in the monomials t_{ij}, $t_{i'j'}$, with $i \neq i'$ and $\lambda(i, j) = -1$, then $\lambda(i', j') = +1$.
(5) If $\lambda(i, j) \neq 0$, then there exists a j' such that $\lambda(i, j') = -1$.
(6) The labeling graph G_λ contains no directed cycles.

We give an equivalent way to understand the definition of a proper labeling.

- ▷ Condition (1) : there is a partition of the polynomial system $\boldsymbol{f}$ into polynomial systems $\boldsymbol{g}$ and $\boldsymbol{h}$ such that all monomials in $\boldsymbol{g}$ are labeled in $\{+1, -1\}$ and all monomials in $\boldsymbol{h}$ are labeled 0.
- ▷ Condition (2) : each polynomial g_i in $\boldsymbol{g}$ can be written as $g_i = g_i^+ + g_i^-$, such that g_i^+ and g_i^- are polynomials on a disjoint set of variables.
- ▷ Condition (3) : Each g_i^- is multilinear.
- ▷ Condition (4) : Any variable x_k can appear in at most one of the g_i^-. Moreover, if an x_k appears in some g_i^-, it does not appear in any h_j in $\boldsymbol{h}$.
- ▷ Condition (5) : Every g_i^- involves at least one variable.
- ▷ Condition (6) : The graph G_λ is essentially between polynomials in $\boldsymbol{g}$ and the variables that appear in them, with an edge $(g_i \to x_k)$ if x_k appears in g_i^- or an edge $(x_k \to g_i)$ if x_k appears in g_i^+.
- ▷ Note that $\deg^{\lambda}(g_i) = \deg(g_i^-)$, whereas $\deg^{\lambda}(h_j) = \deg(h_j)$.

It is easy to see that the trivial labeling $\lambda(i,j) = 0$ is always proper. As we will see this special case of the Labeled CWT corresponds to the original CWT. Note that in this case the labeling graph G_λ is an empty graph. Also, given a system of polynomials $\boldsymbol{f}$ and a labeling λ, it is possible to check in polynomial time whether the labeling λ is proper or not.

Example 2 (continued). It is an instructive exercise to verify that the labeling λ specified was indeed a proper labeling of $\boldsymbol{f}$.

Theorem 30 (Labeled Chevalley-Warning Theorem)**.** *Let $\mathbb{F}_q$ be a finite field with characteristic p and $\boldsymbol{f} \in \mathbb{F}_q[\boldsymbol{x}]^m$. If λ is a proper labeling of $\boldsymbol{f}$ with $\sum_{i=1}^m \deg^\lambda(f_i) < n$, then $|\mathcal{M}_{\boldsymbol{f}}| = 0$. In particular, $|\mathcal{V}_{\boldsymbol{f}}| \equiv 0 \pmod p$.*

Proof. We can re-write $\mathsf{CW}_{\boldsymbol{f}}(x)$ as $\sum_{S\subseteq[m]} \prod_{i\in S} (-1)^{|S|} f_i^{p-1}$. We'll show that every monomial appearing in the expansion of $\prod_{i\in S} f_i^{p-1}$ will have at least one variable with degree at most $p-1$. For simplicity, we focus on the case $S = [m]$ and the other cases of S follow similarly.

We index a monomial of $\prod_{i\in[m]} f_i^{p-1}$ with a tuple $((j_{11}, j_{12}, \ldots, j_{1(p-1)}), \ldots, (j_{m1}, \ldots, j_{m(p-1)}))$ with $1 \le j_{i\ell} \le L_i$ where L_i is the number of monomials in the explicit representation of f_i. The coordinates $(j_{i1}, \ldots, j_{i(p-1)})$ represent the indices of the monomials chosen from each of the $p-1$ copies of f_i^{p-1}. More succinctly, we have $t = \prod_{i=1}^m \prod_{\ell=1}^{p-1} t_{i,j_{i\ell}}$.

Case 1. $\lambda(i, j_{i\ell}) \in \{0, -1\}$, for all (i,ℓ):
Here, $\deg(t) \le (p-1)\sum_{i=1}^m \deg^\lambda(f_i)$ which, by our assumption, is strictly less than $(p-1)n$. Hence, there is a variable with degree less than $p-1$.

Case 2. There is a unique i with $\lambda(i, j_{i\ell}) = +1$ for some ℓ: (warmup for case 3)
That is, for all $i' \neq i$, $\lambda(i', j_{i'\ell}) \in \{0, -1\}$. By condition (5) of proper labeling there exists a $j' \neq j_{i\ell}$ such that $\lambda(i, j') = -1$. Let x_k be a variable in the monomial $t_{ij'}$. By condition (2), x_k is not present in the monomial $t_{i,j_{i\ell}}$ and by condition (3), its degree in $(t_{i,j_{i,1}}, \ldots, t_{i,j_{i,p-1}})$ is at most $p-2$. Additionally, by condition (4), any monomial of $f_{i'}$ for $i' \neq i$ containing x_k must have label $+1$, but $\lambda(i', j_{i',\ell})$ are all in $\{0, -1\}$. Hence, x_k does not appear in any other monomial of t and its degree on t is equal to its degree in $(t_{i,j_{i,1}} \cdots t_{i,j_{i,p-1}})$, which is strictly less than $p-1$.

Case 3. $I = \{i : \lambda(i, j_{i\ell}) = +1$ for some $\ell\}$:
In the labeling graph G_λ, let $i \in I$ be such that there is no path from f_i to any other $f_{i'}$ for $i' \in I$. Such an i exists due to acyclicity of G_λ, i.e. condition (6). Let ℓ be such that $\lambda(i, j_{i\ell}) = +1$. Again, by condition (5) of proper labeling there exists a $j' \neq j_{i\ell}$ such that $\lambda(i, j') = -1$. Let x_k be a variable in the monomial $t_{ij'}$. By condition (2), x_k is not present in the monomial $t_{i,j_{i\ell}}$ and by condition (3), its degree in $(t_{i,j_{i,1}}, \ldots, t_{i,j_{i,p-1}})$ is at most $p-2$. Additionally, by condition (4), any monomial of $f_{i'}$ for $i' \neq i$ containing x_k must have label $+1$. For $i' \notin I$, $\lambda(i', j_{i',\ell})$ are all in $\{0, -1\}$. And for $i' \in I$, variable x_k cannot appear with $+1$ label in $f_{i'}$ by our choice of f_i. Hence, x_k does not appear in any other monomial of t and its degree on t is equal to its degree on $(t_{i,j_{i,1}} \cdots t_{i,j_{i,p-1}})$, which is strictly less than $p-1$.

□

We are now ready to prove the PPA_p-hardness of $\textsc{ChevalleyWithSymmetry}_p$.

Lemma 31. *For all primes p, $\textsc{ChevalleyWithSymmetry}_p$ is PPA_p-hard.*

Proof. We prove that $\textsc{Lonely}_p \preceq \textsc{ChevalleyWithSymmetry}_p$. Let us assume (without loss of generality from Lemma 44) that the $\textsc{Lonely}_p$ instance has a single distinguished vertex represented by 0^n. We'll assume that 0^n is isolated, otherwise, no further reduction is necessary.

Pre-processing. We slightly modify the given circuit $\mathcal{C}$ by defining $\mathcal{C}' : \mathbb{F}_p^n \to \mathbb{F}_p^n$ as follows:

$$\mathcal{C}'(v) = \begin{cases} v & \text{, if } \mathcal{C}^p(v) \neq v \\ \mathcal{C}(v) & \text{, otherwise} \end{cases}$$

Since p is a prime, a vertex $v \in \mathbb{F}_p^n$ has $\deg(v) = 1$ if and only if $\mathcal{C}^p(v) = v$ and $\mathcal{C}(v) \neq v$. By modifying the circuit, we changed this condition to just $\mathcal{C}'(v) \neq v$, which facilitates our reduction.

Circuit $\mathcal{C}'$ is composed of the $\mathbb{F}_p$-addition $(+)$, $\mathbb{F}_p$-multiplication $(\times)$ and the constant (1) gates. However, we require the input of $\textsc{ChevalleyWithSymmetry}_p$ to be a zecote polynomial system, and so we further modify the circuit $\mathcal{C}'$ to eliminate all the constant (1) gates, without changing it's behavior – this is possible because we assume $\mathcal{C}'(0^n) = 0^n$.

Claim 32. *Given circuit $\mathcal{C}'$ with $(+, \times, 1)$ gates, there exists circuit $\bar{\mathcal{C}}$ with $(+, \times)$ gates such that*

$$\bar{\mathcal{C}}(\boldsymbol{v}) = \begin{cases} 0^n & \text{, if } \boldsymbol{v} = 0^n \\ \mathcal{C}'(\boldsymbol{v}) & \text{, otherwise} \end{cases}$$

Proof of Claim 32. We replace all instances of the (1) gate by the function $\mathbb{1}_{\{\boldsymbol{v} \neq 0^n\}}$, which we can compute using only $(+, \times)$ gates as follows: For any $x, y \in \mathbb{F}_p$, observe that $\mathbb{1}_{\{x \neq 0\}} \vee \mathbb{1}_{\{y \neq 0\}} = x^{p-1} + y^{p-1} - x^{p-1}y^{p-1}$. We can thus recursively compute $\bigvee_{i=1}^n \mathbb{1}_{\{v_i \neq 0\}}$ using only $(+, \times)$ gates. Thus, $\bar{\mathcal{C}}(\boldsymbol{v}) = \mathcal{C}'(\boldsymbol{v})$ for all $\boldsymbol{v} \neq 0^n$. And $\bar{\mathcal{C}}(0^n) = 0^n$, since $\bar{\mathcal{C}}$ is computed with only $(+, \times)$ gates. □

Thus, we can transform our original circuit $\mathcal{C}$ into a circuit $\bar{\mathcal{C}}$ with just $(+, \times)$ gates. For simplicity, we'll write $\bar{\mathcal{C}}$ as simply $\mathcal{C}$ from now on.

As an intermiate step in the reduction we describe a system of polynomials $\boldsymbol{f}_{\mathcal{C}}$ over $2n + s$ variables $(x_1, \ldots, x_n, z_1, \ldots, z_s, y_1, \ldots, y_n)$, where s is the size of the circuit $\mathcal{C}$. The variables $\boldsymbol{x} = (x_1, \ldots, x_n)$ correspond to the input of $\mathcal{C}$, the variables $\boldsymbol{y} = (y_1, \ldots, y_n)$ correspond to the output and the variables $\boldsymbol{z} = (z_1, \ldots, z_s)$ correspond to the gates of $\mathcal{C}$. For an addition gate $(+)$ we include a polynomial of the form

$$f(a_1, a_2, a_3) = a_2 + a_3 - a_1,$$

where a_1 is the variable corresponding to the output of the $(+)$ gate and a_2, a_3 are the variables corresponding to its two inputs. Similarly for a multiplication $(\times)$ gate, we include a polynomial of the form

$$f(a_1, a_2, a_3) = a_2 \cdot a_3 - a_1$$

Finally, for the output of the circuit, we include the polynomial

$$f(a, y_i) = a - y_i,$$

where a is the variable corresponding to the i-th output gate of $\mathcal{C}$. It holds that

$$\mathcal{C}(\boldsymbol{x}) = \boldsymbol{y} \quad \Longleftrightarrow \quad \boldsymbol{f}_{\mathcal{C}}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}) = \boldsymbol{0}.$$

We now describe the reduction from an instance of $\textsc{Lonely}_p$ to that of $\textsc{ChevalleyWithSymmetry}_p$. In order to do this, we need to specify a system of polynomials $(\boldsymbol{g}, \boldsymbol{h})$ and a permutation σ such that $|\sigma| = p$. In addition, we will provide a proper labeling λ for $\boldsymbol{g}$ satisfying the degree condition. We will also ensure that $\langle\sigma\rangle$ acts freely on $\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$. And hence, the only valid solutions for the resulting $\textsc{ChevalleyWithSymmetry}_p$ instance will be $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \mathcal{V}_{\boldsymbol{h}}$.

Definition of $\boldsymbol{g}$. The polynomial system $\boldsymbol{g}$ contains the following systems of polynomials.

$$\begin{gathered}
\boldsymbol{f}_{\mathcal{C}}(\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{z}_{1,2}) \\
\boldsymbol{x}_2 - \boldsymbol{x}_3 \\
\boldsymbol{f}_{\mathcal{C}}(\boldsymbol{x}_3, \boldsymbol{x}_4, \boldsymbol{z}_{3,4}) \\
\boldsymbol{x}_4 - \boldsymbol{x}_5 \\
\vdots \\
\boldsymbol{f}_{\mathcal{C}}(\boldsymbol{x}_{2p-1}, \boldsymbol{x}_{2p}, \boldsymbol{z}_{2p-1,2p})
\end{gathered}$$

Note that there are $N = (2n + s)p$ variables in total.

Labeling λ of $\boldsymbol{g}$. For the polynomials belonging to a system of the form $\boldsymbol{f}_{\mathcal{C}}$, the labeling is equal to -1 for the monomials corresponding to the output of each gate and $+1$, otherwise. For instance, let $a_2 + a_3 - a_1$ be the i-th polynomial of $\boldsymbol{g}$ corresponding to a $(+)$ gate and let $a_1 \prec a_2 \prec a_3$, then $\lambda(i, 1) = -1$ and $\lambda(i, 2) = \lambda(i, 3) = +1$.

For the polynomials belonging to a system of the form $\boldsymbol{x}_i - \boldsymbol{x}_{i+1}$, the labeling is equal to -1 for the monomials with variables in $\boldsymbol{x}_{i+1}$ and $+1$ for the monomials with variables in $\boldsymbol{x}_i$.

Claim 33. *The labeling λ for $\boldsymbol{g}$ is proper.*

Proof of Claim 33. By Definition 29, the labeling λ is proper if the following conditions hold.

Condition 1. *For all i, either $\lambda(i, j) \in \{-1, 1\}$ for all j, or $\lambda(i, j) = 0$ for all j.*
In the labeling λ, there are no labels equal to 0, so this condition holds trivially.

Condition 2. *If two monomials t_{ij}, $t_{ij'}$ contain the same variable x_k, then $\lambda(i, j) = \lambda(i, j')$.*
By construction of $\boldsymbol{g}$, no variable appears twice in the same polynomial with a different labeling. For polynomials of $\boldsymbol{f}_{\mathcal{C}}$, this holds because the output variable of a gate is not simultaneously an input variable and all input variables have the same labeling. For polynomials in a system of the form $\boldsymbol{x}_i - \boldsymbol{x}_{i+1}$, each polynomial contains two different variables.

Condition 3. *If $\lambda(i,j) = -1$, then t_{ij} is multilinear.*
For polynomials of $\boldsymbol{f}_{\mathcal{C}}$, only the output variable of a gate has label -1 and by definition this monomial is linear. For polynomials in a system of the form $\boldsymbol{x}_i - \boldsymbol{x}_{i+1}$, all monomials are linear, so the condition holds trivially.

Condition 4. *If x_k is a variable in the monomials t_{ij}, $t_{i'j'}$, with $i \neq i'$ and $\lambda(i,j) = -1$, then $\lambda(i',j') = +1$.*
Observe that all monomials with label -1 contain only a single variable, so we refer to a monomial x_k with label -1. For a polynomial in $\boldsymbol{f}_{\mathcal{C}}$, a monomial x_k with label -1 corresponds to the output of a gate. Hence, if x_k appears in other monomials of $\boldsymbol{f}_{\mathcal{C}}$, these monomials correspond to inputs and have label $+1$. Also, if x_k is an output variable of $\boldsymbol{f}_{\mathcal{C}}$, then it might appear in a polynomial of the form $a_1 - a_2$. However, by construction the monomials of $x_i - x_{i+1}$ that correspond to output variables of $\boldsymbol{f}_{\mathcal{C}}$ have label $+1$.

Condition 5. *If $\lambda(i,j) \neq 0$, then there exists a j' such that $\lambda(i,j') = -1$.*
By the definition of λ, all polynomials of $\boldsymbol{g}$ have a monomial with label -1. These are the monomials that correspond to the outputs of a gate for the systems of the form $\boldsymbol{f}_{\mathcal{C}}$ and the monomials that correspond to $\boldsymbol{x}_{i+1}$ for the systems of the form $\boldsymbol{x}_i - \boldsymbol{x}_{i+1}$.

Condition 6. *The labeling graph G_λ contains no cycles.*
Each system of the form $\boldsymbol{x}_i - \boldsymbol{x}_{i+1}$ has incoming edges with variables appearing only in the i-th copy of $\boldsymbol{f}_{\mathcal{C}}$ and outgoing edges with variables appearing only in the $(i+1)$-th copy of $\boldsymbol{f}_{\mathcal{C}}$. Also, the variables appearing on the i-th copy of $\boldsymbol{f}_{\mathcal{C}}$ might appear only in the systems $\boldsymbol{x}_{i-1} - \boldsymbol{x}_i$ and $\boldsymbol{x}_i - \boldsymbol{x}_{i+1}$. Hence, G_λ has no cycles that contain vertices of two different copies of $\boldsymbol{f}_{\mathcal{C}}$ or of a copy of $\boldsymbol{f}_{\mathcal{C}}$ and a system of the form $\boldsymbol{x}_{i-1} - \boldsymbol{x}_i$.

It is left to argue that the labeling graph restricted to a copy of $\boldsymbol{f}_{\mathcal{C}}$ does not have any cycles. Let the vertices of $\boldsymbol{f}_{\mathcal{C}}$ be ordered according to the topological ordering of $\mathcal{C}$. This restricted part of G_λ corresponds exactly to the graph of $\mathcal{C}$, which by definition is a DAG. Hence, G_λ contains no cycles. □

We also need to show that for this labeling $\boldsymbol{g}$ satisfies the labeled Chevalley condition.

Claim 34. *The labeled Chevalley condition $\sum_{i=1}^{m_g} \deg^{\lambda}(g_i) < N$ holds for $\boldsymbol{g}$ with labeling λ.*

Proof. Each polynomial of $\boldsymbol{g}$ has a unique monomial with $\lambda(i,j) = -1$ and this monomial has degree 1. Thus, $\sum_{i=1}^{m_g} \deg^{\lambda}(g_i) = m_g$. On the other hand, the i-th polynomial of $\boldsymbol{g}$ has exactly one variable that has not appeared in any of the previous polynomials. More specifically, the number of variables is equal to $m_g + n$, where n is the size of the input of $\mathcal{C}$. Hence, the labeled Chevalley condition holds for $\boldsymbol{g}$. □

Definition of $\boldsymbol{h}$. The system of polynomials $\boldsymbol{g}$ allows us to compute the p vertices given by $\mathcal{C}^i(\boldsymbol{x})$ for $i \in [p+1]$. From the definition of $\textsc{Lonely}_p$ and our pre-processing on $\mathcal{C}$, this group of p vertices is a hyperedge if and only if

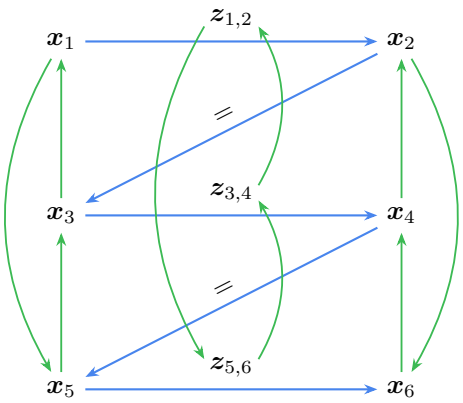
$\mathcal{C}(\boldsymbol{x}) \neq \boldsymbol{x}$. Since solutions of $\textsc{Lonely}_p$ are lonely vertices, we define $\boldsymbol{h}$ to exclude $\boldsymbol{x}$ such that $\mathcal{C}(\boldsymbol{x}) \neq \boldsymbol{x}$. Namely, we set $\boldsymbol{h}$ to be the system of polynomials

$$\boldsymbol{x}_1 - \boldsymbol{x}_2.$$

Definition of permutation σ. In the description of $\boldsymbol{f} = (\boldsymbol{g}, \boldsymbol{h})$, we have used the following vector of variables:

$$\boldsymbol{x} = (\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_{2p}, \boldsymbol{z}_{1,2}, \boldsymbol{z}_{3,4}, \ldots, \boldsymbol{z}_{2p-1,2p})$$

We define the permutation σ such that $\sigma(\boldsymbol{x}) = (\boldsymbol{x}_3, \boldsymbol{x}_4, \ldots, \boldsymbol{x}_{2p}, \boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{z}_{3,4}, \boldsymbol{z}_{5,6}, \ldots, \boldsymbol{z}_{2p-1,2p}, \boldsymbol{z}_{1,2})$, as illustrated in the following figure. The blue arrows indicate the polynomials $\boldsymbol{g}$ and the green arrows indicate the permutation σ in the case of $p = 3$.



Claim 35. *The group $\langle\sigma\rangle$ has order p and acts freely on $\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$.*

Proof. In order to see that $|\sigma| = p$, note that the input of σ consists of $3p$ blocks of variables. The permutation σ performs a rotation of the first $2p$ blocks by two positions and of the last p blocks by one position. All that remains is to show that $\langle\sigma\rangle$ acts freely on $\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$. First, we show that $\langle\sigma\rangle$ defines a group action on $\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$, that is for all $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$, it holds that $\sigma(\boldsymbol{x}) \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$. Let $\boldsymbol{x} = (\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_{2p-1}, \boldsymbol{x}_{2p}, \boldsymbol{z}_{1,2}, \boldsymbol{z}_{3,4}, \ldots, \boldsymbol{z}_{2p-1,2p}) \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$, then

- $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}}$ implies that $\boldsymbol{f}_{\mathcal{C}}(\boldsymbol{x}_{2i-1}, \boldsymbol{x}_{2i}, \boldsymbol{z}_{2i-1,2i}) = 0$ for $i \in [p]$ and $\boldsymbol{x}_{2i} = \boldsymbol{x}_{2i+1}$ for $i \in [p-1]$
- $\boldsymbol{x} \in \overline{\mathcal{V}_{\boldsymbol{h}}}$ implies that $\boldsymbol{x}_1 \neq \boldsymbol{x}_2$, that is, $\mathcal{C}(\boldsymbol{x}_1) \neq \boldsymbol{x}_1$ since $\boldsymbol{f}_{\mathcal{C}}(\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{z}_{1,2}) = 0 \Leftrightarrow \boldsymbol{x}_2 = \mathcal{C}(\boldsymbol{x}_1)$.

Now, $\sigma(\boldsymbol{x}) = (\boldsymbol{x}_3, \boldsymbol{x}_4, \ldots, \boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{z}_{3,4}, \boldsymbol{z}_{5,6}, \ldots, \boldsymbol{z}_{1,2}) \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$ holds because

- $\boldsymbol{f}_{\mathcal{C}}(\boldsymbol{x}_{2i-1}, \boldsymbol{x}_{2i}, \boldsymbol{z}_{2i-1,2i}) = 0$ for $i \in [p]$ and $\boldsymbol{x}_{2i} = \boldsymbol{x}_{2i+1}$ for $i \in [p-1]$, which holds from $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}}$. Additionally, $\boldsymbol{x}_1 = \boldsymbol{x}_{2p}$ holds because we pre-processed $\mathcal{C}$ such that $\mathcal{C}^p(\boldsymbol{x}_1) = \boldsymbol{x}_1$,
- $\boldsymbol{x}_3 \neq \boldsymbol{x}_4$, which holds because $\boldsymbol{x}_4 = \mathcal{C}(\boldsymbol{x}_3)$ for $i \in [p]$ and from the definition of $\mathcal{C}$, $\mathcal{C}(\boldsymbol{x}_1) \neq \boldsymbol{x}_1$ implies that $\boldsymbol{x}_{2i} \neq \boldsymbol{x}_{2i-1}$ for all $i \in [p]$.

Finally, if $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$, by construction of $\mathcal{C}$, we have that $\boldsymbol{x}_{2k} \neq \boldsymbol{x}_{2j}$ for $k \neq j$ and thus $\sigma(\boldsymbol{x}) \neq \boldsymbol{x}$ simply because $\boldsymbol{x}_3 \neq \boldsymbol{x}_1$. Thus, we conclude that $\langle\sigma\rangle$ acts freely on $\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$. □

Putting it all together. The solution of this instance of ChevalleyWithSymmetry$_p$ cannot be a vector $\boldsymbol{x} \in \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$ with $\sigma(\boldsymbol{x}) \notin \mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$ or $\sigma(\boldsymbol{x}) = \boldsymbol{x}$, since we know from Claim 35 that $\langle\sigma\rangle$ acts freely on $\mathcal{V}_{\boldsymbol{g}} \cap \overline{\mathcal{V}_{\boldsymbol{h}}}$. We also have from Theorem 30 that the solution also cannot be a max-degree monomial in the expansion of $\mathsf{CW}_{\boldsymbol{g}}(\boldsymbol{x}) = \prod(1 - g_i^{p-1})$. Thus, the solution must be an $\boldsymbol{x} \neq \boldsymbol{0}$ such that $\boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{0}$. Let $\boldsymbol{x}_1$ denote the first n coordinates of $\boldsymbol{x}$, then $\boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{0}$ implies that $\boldsymbol{x}_1 = \mathcal{C}(\boldsymbol{x}_1)$ and $\boldsymbol{x} \neq \boldsymbol{0}$ implies that $\boldsymbol{x}_1 \neq \boldsymbol{0}$. Hence, $\boldsymbol{x}_1$ corresponds to a lonely vertex of the Lonely$_p$ instance. □

Required restricted-Lonely support

Lemma 44. *For all primes p and $k \in \{1, \ldots, p-1\}$, Lonely$_p$ reduces to Lonely$_p^k$.*

A Appendix: Reductions Between Complete Problems

In order to prove Theorem 9, we introduce an additional problem that will serve as intermediate problem in our reductions.

Definition 47. (Leaf$'_q$)

Principle: Same as Leaf$_q$, but degrees are allowed to be larger (polynomially bounded).

Object: q-uniform hypergraph $G = (V, E)$. Designated vertex $v^* \in V$.

Inputs: ▷ $C : \{0,1\}^n \to (\{0,1\}^{nq})^k$, with $(\{0,1\}^{nq})^k$ interpreted as k many q-subsets of $\{0,1\}^n$
▷ $v^* \in \{0,1\}^n$ (usually 0^n)

Encoding: $V \coloneqq \{0,1\}^n$.
For distinct $v_1, \ldots, v_q$, edge $e \coloneqq \{v_1, \ldots, v_q\} \in E$ if $e \in C(v)$ for all $v \in e$

Solutions: v^* if $\deg(v) \equiv 0 \pmod q$ and
$v \neq v^*$ if $\deg(v) \not\equiv 0 \pmod q$

Proof of Theorem 9. We show the following inter-reducibilities: (1) Leaf$_q \asymp$ Leaf$'_q$, (2) Leaf$'_q \asymp$ Bipartite$_q$ and (3) Leaf$_q \asymp$ Lonely$_q$.

(1a) Leaf$_{\boldsymbol{q}} \preceq$ Leaf$'_{\boldsymbol{q}}$ Each instance of Leaf$_q$ is trivially an instance of Leaf$'_q$.

(1b) Leaf$'_{\boldsymbol{q}} \preceq$ Leaf$_{\boldsymbol{q}}$. We start with a Leaf$'_q$ instance (C, v^*), where C encode a q-uniform hypergraph $G = (V, E)$ with degree at most k. Let $t = \lceil k/q \rceil$. We construct a Leaf$_q$ instance encoding a hypergraph $\overline{G} = (\overline{V}, \overline{E})$ on vertex set $\overline{V} \coloneqq V \times [t]$, intuitively making t copies of each vertex.

In order to locally compute hyperedges, we first fix a canonical algorithm that for any vertex v and any edge $e \in E$ incident on v, assigns it a label $\ell_v(e) \in [t]$, with at most q edges mapping to the same label — e.g. sort all edges incident on v in lexicographic order and bucket them sequentially in at

most t groups of at most q each. Note that we can ensure that for any vertex v at most one label gets mapped to by a non-zero, non-q number of edges. Moreover, if $\deg(v) \equiv 0 \pmod q$, then exactly q or 0 edges are assigned to any label.

We'll assume that $\deg(v^*) \not\equiv 0 \pmod q$, as otherwise, a reduction wouldn't be necessary. We let (v^*, ℓ^*) be the designated vertex of the $\textsc{Leaf}_q$ instance, where ℓ^* is the unique label that gets mapped to by a non-zero, non-q number of edges incident on v^*.

For any vertex $(v, i) \in \overline{V}$, we assign it at most q edges as follows: For each edge $e = \{v_1, \ldots, v_q\}$ such that $\ell_v(e) = i$, the corresponding hyperedge of (v, i) is $(v_1, \ell_{v_1}(e)), \ldots, (v_q, \ell_{v_q}(e))$. It is easy to see that the designated vertex (v^*, ℓ^*) indeed has non-zero, non-q degree. Moreover, a vertex $\deg(v, i) \notin \{0, q\}$ in $\overline{G}$ only if v has a non-multiple-of-q degree in G. Thus, solutions to the $\textsc{Leaf}_q$ instance naturally maps to solutions to the original $\textsc{Leaf}'_q$ instance.

By Remark 10, this completes the reduction since the edges are locally computable with black-box access to C and $\overline{V}$ is efficiently indexable.

(2a) $\textsc{Leaf}'_{\boldsymbol{q}} \preceq \textsc{Bipartite}_{\boldsymbol{q}}$. We start with a $\textsc{Leaf}'_q$ instance (C, v^*), where C encode a q-uniform hypergraph $G = (V, E)$. We construct a $\textsc{Bipartite}_q$ instance encoding a graph $\overline{G} = (\overline{V} \cup \overline{U}, \overline{E})$ such that $\overline{V} = V$ and $\overline{U} = \binom{V}{q}$, i.e. all q-sized subsets of V. We include the edge $(v, e) \in \overline{E}$ if $e \in E$ is incident on v. The designated vertex for the $\textsc{Bipartite}_q$ instance is v^* in $\overline{V}$.

Clearly, all vertices $e \in \overline{U}$ have degree either q or 0. For any $v \in \overline{V}$, the degree of v in $\overline{G}$ is same as its degree in G. Thus, any solution to the $\textsc{Bipartite}_q$ instance immediately gives a solution to the original $\textsc{Leaf}'_q$ instance. By Remark 10, this completes the reduction since the edges are locally computable with black-box access to C and $\overline{V}$ and $\overline{U}$ are efficiently indexable (cf. [31, §2.3] for efficiently indexing $\overline{U}$).

(2b) $\textsc{Bipartite}_{\boldsymbol{q}} \preceq \textsc{Leaf}'_{\boldsymbol{q}}$. We start with a $\textsc{Bipartite}_q$ instance (C, v^*) encoding a bipartite graph $\mathcal{G} = (V \cup U, E)$ with maximum degree of any vertex being at most k. We construct a $\textsc{Leaf}'_q$ instance encoding a hypergraph $\overline{G} = (\overline{V}, \overline{E})$ such that $\overline{V} = V$ with designated vertex v^*.

First, we fix a canonical algorithm that for any vertex $u \in U$ with $\deg_G(u) \equiv 0 \pmod q$ produces a partition of it's neighbors with q vertices of V in each part. Now, the set of q-uniform hyperedges incident on any vertex $v \in \overline{V}$ in $\overline{E}$ can be obtained as: for all neighbors u of v, with $\deg_G(u) \equiv 0 \pmod q$, we include a hyperedge consisting of all vertices in the same partition as v among the neighbors of u (we ignore neighbors u with $\deg(u) \not\equiv 0 \pmod q$).

Observe that $\deg_{\overline{G}}(v) \leq \deg_G(v)$ and equality holds if and only if all neighbors of v in G have degree $\equiv 0 \pmod q$. Hence for any $v \in \overline{V}$, if $\deg_{\overline{G}}(v) \neq \deg_G(v) \pmod q$, then there exists a neighbor $u \in U$ of v in G such that $\deg(u) \not\equiv 0 \pmod q$. Thus, if $v = v^*$ and $\deg_{\overline{G}}(v^*) \equiv 0 \pmod q$, then either $\deg_G(v) \equiv 0 \pmod q$ or we can find a neighbor u of v in G with $\deg(u) \not\equiv 0 \pmod q$. Similarly if for some $v \neq v^*$, we have $\deg_{\overline{G}}(v^*) \not\equiv 0 \pmod q$, then either $\deg_G(v) \not\equiv 0 \pmod q$ or we can find a neighbor u of v in G with $\deg(u) \not\equiv 0 \pmod q$. Thus, any solution to the $\textsc{Leaf}'_q$ instance gives us a solution to the original $\textsc{Bipartite}_q$ instance. This completes the reduction since $\overline{V} = \{0, 1\}^n$ and the edges are locally computable with black-box access to C.

(3a) $\textsc{Leaf}_{\boldsymbol{q}} \preceq \textsc{Lonely}_{\boldsymbol{q}}$. We start with a $\textsc{Leaf}_q$ instance (C, v^*), where C

encode a q-uniform hypergraph $G = (V, E)$ with degree at most q. If $\deg_G(v^*) = q$ or 0, then we don't need any further reduction. Else, we construct a $\textsc{Lonely}_q$ instance encoding a q-dimensional matching $\overline{G} = (\overline{V}, \overline{E})$ on vertex set $\overline{V} = V \times [q]$. The designated vertices will be $V^* = \{(v, q - i) : 1 \le i \le q - \deg(v^*)\}$. Note that, $|V^*| = q - \deg_G(v^*)$ and hence $1 \le |V^*| \le q - 1$.

In order to locally compute hyperedges, we first fix a canonical algorithm that for any vertex v and any edge $e \in E$ incident on v, assigns it a unique label $\ell_v(e) \in [q]$ — e.g. sort all edges incident on v in lexicographic order and label them sequentially in $[q]$. In fact, we can ensure that an edge incident on v get labeled within $\{1, \ldots, \deg_G(v)\}$.

For any vertex $(v, i) \in \overline{V}$, we assign it at most one hyperedge as follows:

- ▷ If $\deg_G(v) = 0$, we include the hyperedge $\{(v, i) : i \in [q]\}$.
- ▷ Else if $\deg_G(v) \ge i$, then for edge $e = \{v_1, \ldots, v_q\}$ incident on v such that $\ell_v(e) = i$, the corresponding hyperedge of (v, i) is $(v_1, \ell_{v_1}(e)), \ldots, (v_q, \ell_{v_q}(e))$.
- ▷ Else if $0 < \deg_G(v) < i$, we leave it isolated.

It is easy to see that our definition of hyperedges is consistent and that the designated vertices V^* are indeed isolated. Moreover, a vertex (v, i) is isolated in $\overline{G}$ only if $1 \le \deg_G(v) \le q - 1$. Thus, solutions to the $\textsc{Leaf}_q$ instance naturally maps to solutions to the original $\textsc{Leaf}'_q$ instance.

By Remark 10, this completes the reduction since the edges are locally computable with black-box access to C and $\overline{V}$ is efficiently indexable.

(3b) $\textsc{Lonely}_{\boldsymbol{q}} \preceq \textsc{Leaf}_{\boldsymbol{q}}$. We start with a $\textsc{Lonely}_q$ instance (C, V^*), where C encode a q-dimensional matching $G = (V, E)$. We construct a $\textsc{Leaf}_q$ instance encoding a q-uniform hypergraph $\overline{G} = (\overline{V}, \overline{E})$ on vertex set $\overline{V}$ that will be specified shortly. We describe the hyperedges in $\overline{G}$ and it'll be clear how to compute the hyperedges for any vertex locally with just black-box access to C.

We start with $\overline{V} = V$. Our goal is to transform all vertices of degree 1 to degree q, while ensuring that vertices of degree 0 are mapped to vertices of degree not a multiple of q. Towards this goal we let $\overline{E}$ to be set of edges in E in addition to $q - 1$ canonical q-dimensional matchings over V. For example, for a vertex $v \coloneqq (x_1, \ldots, x_n) \in V = [q]^n$, the corresponding edges in $\overline{E}$ include an edge in E (if any) and edges of the type $e_i = \{(x_1, \ldots, x_{i-1}, j, x_{i+1}, \ldots, x_n) : j \in [q]\}$ for $i \in [q - 1]$ (note, this requires us to assume $n \ge q - 1$). Adding the $q - 1$ matchings increases the degree of each vertex by $q - 1$. Therefore, vertices with initial degree 1 now have degree q and vertices with initial degree 0 now have degree $q - 1$. However, a couple of issues remain in order to complete the reduction, which we handle next.

Multiplicities. An edge $e \in E$ might have gotten added twice, if it belonged to one of the canonical matchings. To avoid this issue altogether, instead of adding edges directly on V, we augment $\overline{V}$ to become $\overline{V} \coloneqq V \cup \left(\binom{V}{q} \times [q - 1]\right)$, i.e. in addition to V, we have $q - 1$ vertices for every potential hyperedge of G. For any edge $e \coloneqq \{v_1, \ldots, v_q\} \in E$, instead of adding it directly in $\overline{G}$, we add hyperedge $\{v, (e, 1), (e, 2), \ldots, (e, q - 1)\}$ for each $v \in e$. Note that, all vertices $(e, i) \in \binom{V}{q} \times [q - 1]$ have degree q if $e \in E$ and degree 0 if $e \notin E$, so they are non-solutions for the $\textsc{Leaf}_q$ instance. For vertices in V, we still have as before that vertices with initial degree 1 now have degree q and vertices with initial degree 0 now have degree $q - 1$.

Designated vertex. In a $\textsc{Leaf}_q$ instance, we need to specify a single designated vertex $v^* \in \overline{V}$. If the $\textsc{Lonely}_q$ instance had a single designated vertex then we would be done. However, in general it is not possible to assume this (for non-prime q). Nevertheless, we provide a way to get around this. We augment $\overline{V}$ with $t = (q-1)(q-k)+1$ additional vertices to become $\overline{V} \coloneqq V \cup \left(\binom{V}{q} \times [q-1]\right) \cup \{w_{i,j} : i \in [q-k],\ j \in [q-1]\} \cup \{v^*\}$, where v^* will eventually be the single designated vertex for the $\textsc{Leaf}_q$ instance.

Let $V^* = \{u_1, \ldots, u_k\} \subseteq V$ be the set of designated vertices in the $\textsc{Lonely}_q$ instance (note $1 \leq k < q$). So far, note that $\deg_{\overline{G}}(u_i) = q-1$. The only new hyperedges we add will be among u_i's, $w_{i,j}$'s and v^*, in such a way that $\deg_{\overline{G}}(u_i)$ will become q, the degree of all $w_{i,j}$'s will also be q and degree of v^* will be $q-k$.

- ▷ For each $u \in V^*$, include $\{u, w_{1,1}, \ldots, w_{1,q-1}\}$. So far, $\deg_{\overline{G}}(u) = q$ and $\deg_{\overline{G}}(w_{1,j}) = k$.
- ▷ For each $j \in [q-1]$ and each $i \in \{2, \ldots, q-k\}$, include $\{w_{1,j}, w_{i,1}, \ldots, w_{i,q-1}\}$. So far, $\deg_{\overline{G}}(w_{i,j}) = q-1$ for all $(i,j) \in [q-k] \times [q-1]$.
- ▷ Finally, for each $(i,j) \in [q-k] \times [q-1]$, include $\{v^*, w_{i,1}, \ldots, w_{i,q-1}\}$. Now, $\deg_{\overline{G}}(w_{i,j}) = q$ for all $(i,j) \in [q-k] \times [q-1]$ and $\deg_{\overline{G}}(v^*) = q-k$.

Thus, we have finally reduced to a $\textsc{Leaf}_q$ instance encoding the graph $\overline{G} = (\overline{V}, \overline{E})$ with $\overline{V} \coloneqq V \cup \left(\binom{V}{q} \times [q-1]\right) \cup \{w_{i,j} : i \in [q-k],\ j \in [q-1]\} \cup \{v^*\}$. By Remark 10, this completes the reduction, since $\overline{V}$ is efficiently indexable (again, see [31] for a reference on indexing $\binom{V}{q}$) and the edges are locally computable using black-box access to C. □

A.1 Completeness of Succinct Bipartite

We introduce a new intermediate problem to show PPA_p–completeness of $\textsc{SuccinctBipartite}_p$.

Definition 48. ($\textsc{TwoMatchings}_p$)

Principle: Two p-dimensional matchings over a common vertex set, with a vertex in exactly one of the matchings, has another such vertex.

Object: Two p-dimensional matchings $G_0 = (V, E_0)$, $G_1 = (V, E_1)$. Designated vertex $v^* \in V$.

Inputs: ▷ $C_0 : \{0,1\}^n \to (\{0,1\}^n)^p$ and $C_1 : \{0,1\}^n \to (\{0,1\}^n)^p$
▷ $v^* \in \{0,1\}^n$

Encoding: $V \coloneqq \{0,1\}^n$. For $b \in \{0,1\}$, $E_b \coloneqq \{e : C_b(v) = e \text{ for all } v \in e\}$

Solutions: v^* if $\deg_{G_0}(v^*) \neq 1$ or $\deg_{G_1}(v^*) \neq 0$ and
$v \neq v^*$ if $\deg_{G_0}(v^*) \neq \deg_{G_1}(v^*)$

Observe that in the case of $p = 2$, $\textsc{TwoMatchings}_p$ can be readily seen as equivalent to $\textsc{Leaf}_2$.

Theorem 49. *For any prime p, $\textsc{SuccinctBipartite}_p$ and $\textsc{TwoMatchings}_p$ are PPA_p–complete.*

Proof. We show that $\textsc{Bipartite}_p \preceq \textsc{SuccinctBipartite}_p \preceq \textsc{TwoMatchings}_p \preceq \textsc{Lonely}_p$.

$\textsc{Bipartite}_{\boldsymbol{p}} \preceq \textsc{SuccinctBipartite}_{\boldsymbol{p}}$. Since p is a prime, we can assume that the designated vertex v^* has degree 1 (mod p) (similar to Lemma 44). Since the number of neighbors in a $\textsc{Bipartite}_p$ instance are polynomial, we can check if an edge exists and canonically group them efficiently for all vertices with degree being a multiple of p. The designated edge e^* is the unique ungrouped edge incident on v^*. Thus, valid solution edges to $\textsc{SuccinctBipartite}_p$ must have at least one endpoint which is a solution to the original $\textsc{Bipartite}_p$ instance.

$\textsc{SuccinctBipartite}_{\boldsymbol{p}} \preceq \textsc{TwoMatchings}_{\boldsymbol{p}}$. We reduce to a $\textsc{TwoMatchings}_p$ instance encoding two p-dimensional matchings $\overline{G}_0 = (\overline{V}, \overline{E}_0)$ and $\overline{G}_1 = (\overline{V}, \overline{E}_1)$, over the vertex set $\overline{V} = V \times U \times [p-1]$, that is, all possible edges producible in the $\textsc{SuccinctBipartite}_p$ instance. The designated vertex v^* is the designated edge e^* in the $\textsc{SuccinctBipartite}_p$ instance.

For any edges $e_1, \ldots, e_p$, which are grouped by ϕ_V pivoted at some $v \in V$, we include the hyperedge $\{e_1, \ldots, e_p\}$ in $\overline{E}_0$. Similarly, for any edges $e_1, \ldots, e_p$, which are grouped by ϕ_U pivoted at some $u \in U$, we include the hyperedge $\{e_1, \ldots, e_p\}$ in $\overline{E}_1$. It is easy to see that points in exactly one of the two matchings $\overline{G}_0$ or $\overline{G}_1$ correspond to edges of the $\textsc{SuccinctBipartite}_p$ instance that are not grouped at exactly one end. Thus, we can derive a solution to $\textsc{SuccinctBipartite}_p$ from a solution to $\textsc{TwoMatchings}_p$. (Remark: while edges which are not grouped at either end are solutions to $\textsc{SuccinctBipartite}_p$, they do not correspond to a solution in the $\textsc{TwoMatchings}_p$ instance.)

$\textsc{TwoMatchings}_p \preceq \textsc{Lonely}_p$. Given a $\textsc{TwoMatchings}_p$ instance encoding two p-dimensional matchings $G_0 = (V, E_0)$ and $G_1 = (V, E_1)$, we reduce to an instance of $\textsc{Lonely}_p$ encoding a p-dimensional matching $\overline{G} = (\overline{V}, \overline{E})$ such that $\overline{V} = V \times [p]$. The designated vertex for the $\textsc{Lonely}_p$ instance is (v^*, p).

For any hyperedge $\{v_1, \ldots, v_p\}$ in E_0, we include the hyperedge $\{(v_1, i), (v_2, i), \ldots, (v_p, i)\}$ in $\overline{G}$ for each $i \in \{1, \ldots, p-1\}$. Similarly, for any hyperedge $\{v_1, \ldots, v_p\}$ in E_1, we include the hyperedge $\{(v_1, p), (v_2, p), \ldots, (v_p, p)\}$ in $\overline{G}$. If $v \in V$ is isolated in both G_0 and G_1, then we include the hyperedge $\{v\} \times [p]$.

Observe that, (v^*, p) is isolated by design. A vertex (v, i), for $i < p$ is isolated only if $\deg_{G_0}(v) = 0$ and $\deg(G_1) = 1$. Similarly, the vertex (v, p) is isolated only if $\deg_{G_0}(v) = 1$ and $\deg(G_1) = 0$. Thus, isolated vertices in the $\textsc{Lonely}_p$ instance correspond to solutions of the $\textsc{TwoMatchings}_p$ instance. □

A.2 Restricted-Lonely support

Proof of Lemma 44. Consider any prime p. Consider a $\textsc{Lonely}_p$ instance (C, V^*), where C encodes a p-dimensional matching $G = (V, E)$ and $|V^*| = \ell$. We wish to reduce to an instance of $\textsc{Lonely}_p^k$, where the number of designated vertices is exactly k. First, we'll assume that all vertices in V^* are indeed isolated in G, otherwise, no reduction would be necessary. The key reason why this lemma holds for primes (and not for composites) is because ℓ has a multiplicative inverse modulo p. In particular, let $t \equiv \ell^{-1} k \pmod{p}$.

We construct a $\textsc{Lonely}_p^k$ instance encoding the p-dimensional matching $\overline{G} = (\overline{V}, \overline{E})$ over $\overline{V} = V \times [t]$. We let $\overline{V}^*$ to be the lexicographically first k vertices in $V^* \times [t]$. Note that $|V^* \times [t]| = t.\ell \equiv k \pmod{p}$. Thus, we partition the remaining vertices of $V^* \times [t]$ into p-uniform hyperedges. For any vertex $v \in V \setminus V^*$, with neighbors $v_1, \ldots, v_{p-1}$ in G, the neighbors of (v, i) in $\overline{G}$ are $(v_1, i), \ldots, (v_{p-1}, i)$ for any $i \in [t]$. Thus, a vertex (v, i) is isolated only if it is

in $\overline{V}^*$ or v is isolated in G. This completes the reduction since $\overline{V}$ is efficiently indexable – see Remark 10. □

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